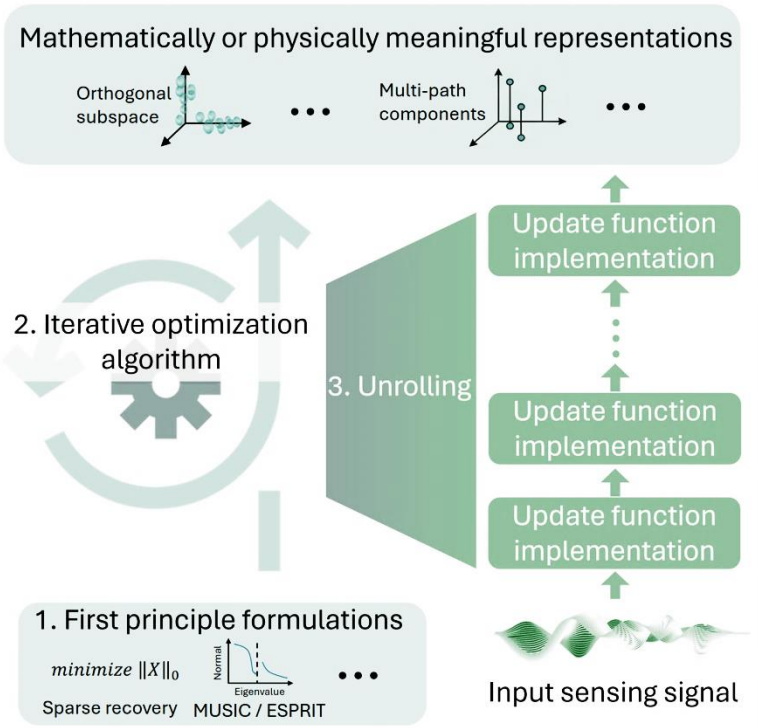
Towards White-Box Deep Wireless Sensing

RF-CRATE: a complex-valued transformer derived from sparse rate reduction

Xie Zhang[1] · Yina Wang[2] · Chenshu Wu[1]

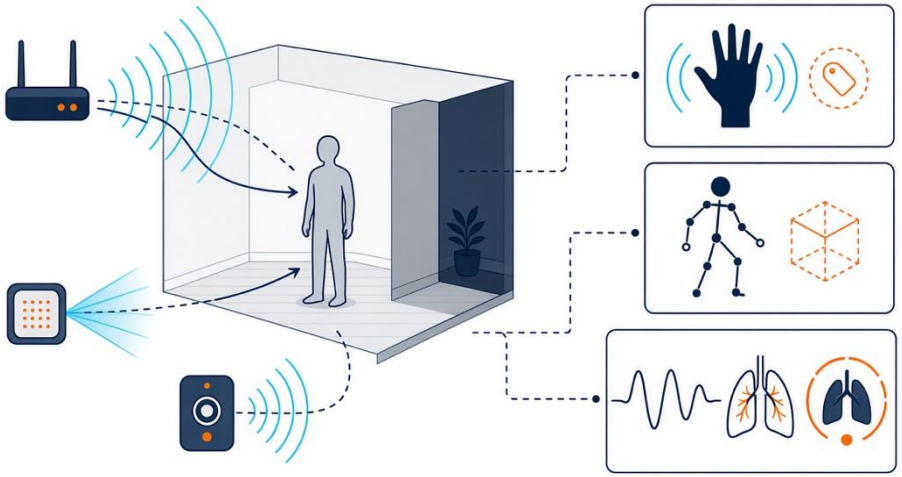
[1]The University of Hong Kong, Hong Kong
[2]MIT, USA



香港大學 THE UNIVERSITY OF HONG KONG | HKU AIoT LAB 香港大學人工智能物聯網實驗室

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Why deep wireless sensing matters



RF signals enable device-free human sensing

- Activity and gesture recognition
- Gait identification
- 2D and 3D pose estimation
- Respiration monitoring

The empirical success of deep learning has spurred its application to the radio-frequency (RF) domain, leading to significant advances in Deep Wireless Sensing (DWS).

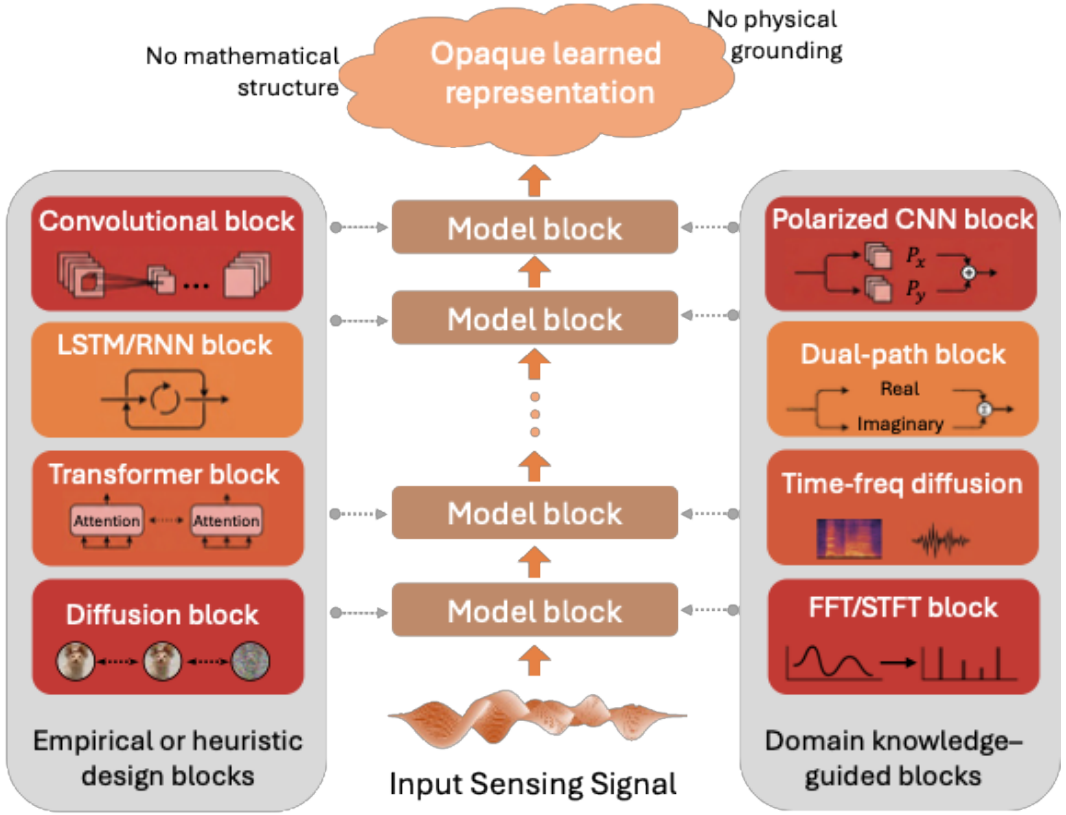
The problem: today's DWS models are mostly black boxes

Although domain knowledge may inform individual components, most existing DWS models remain black boxes with limited interpretability.

Impedes domain knowledge transfer across tasks and modalities.

Lacks generalizability to unseen environments or out-of-distribution conditions.

Raises trust concerns in security-sensitive and safety-critical deployments.



What is “white-box” models

Black-box model

Architecture chosen empirically

↓

Representations learned implicitly

↓

Interpretation added after training

White-box model

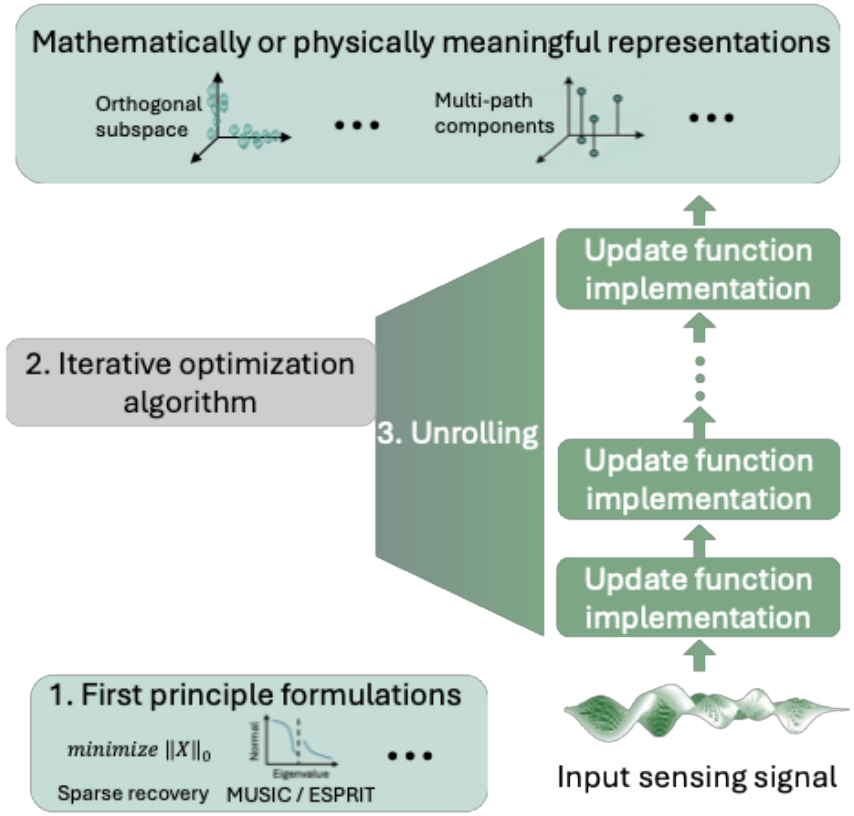
Explicit mathematical/physical objective

↓

Iterative optimization algorithm

↓

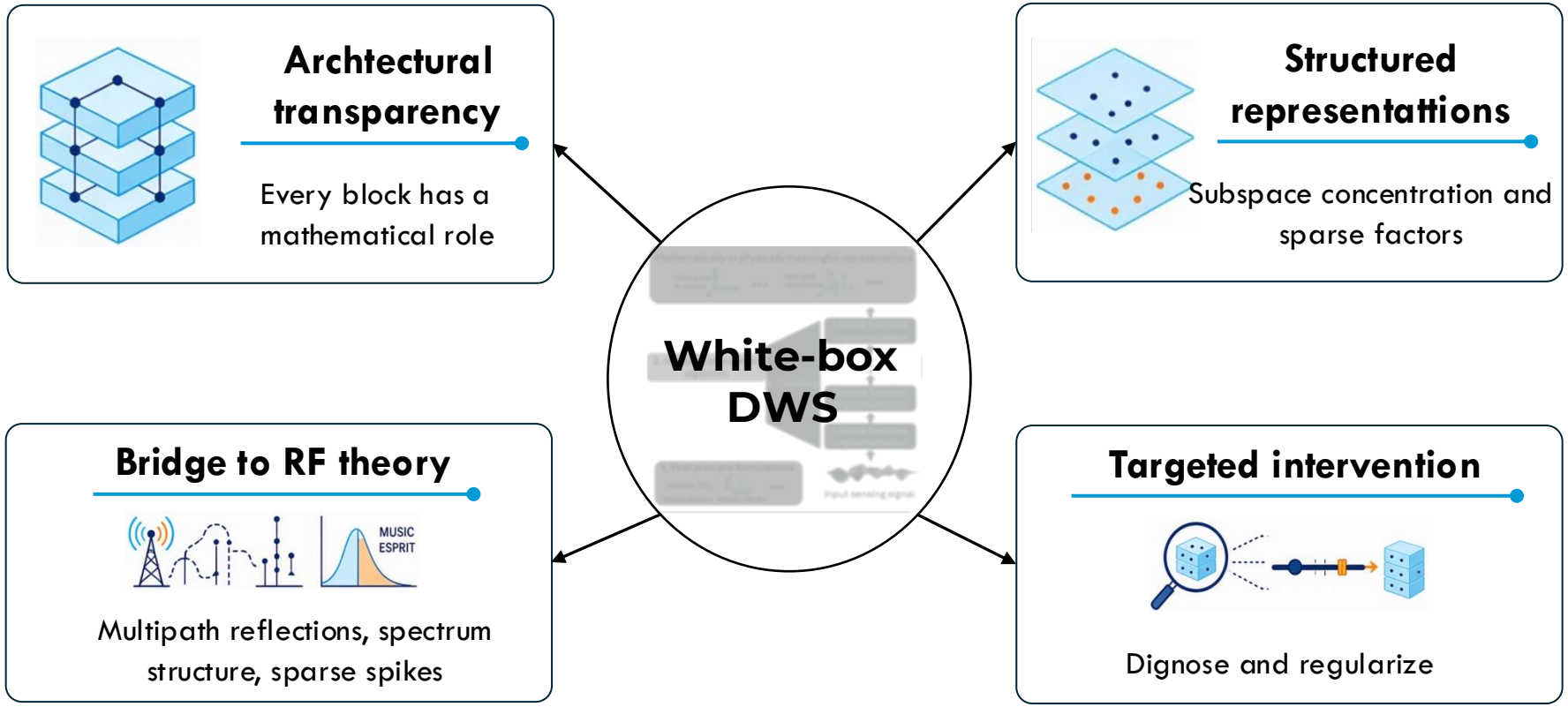
Unrolling the iterations yields the Model



White-box Model

Each module corresponds to a defined optimization step.

Why white-box DWS is worth pursuing



Key observation: RF sensing is parsimonious

WiFi-based gesture recognition as an example:



High-Dimensional Inputs:
2 seconds CSI × 30 subcarriers × 3 links × 1000 Hz
= 180,000 input dimensions

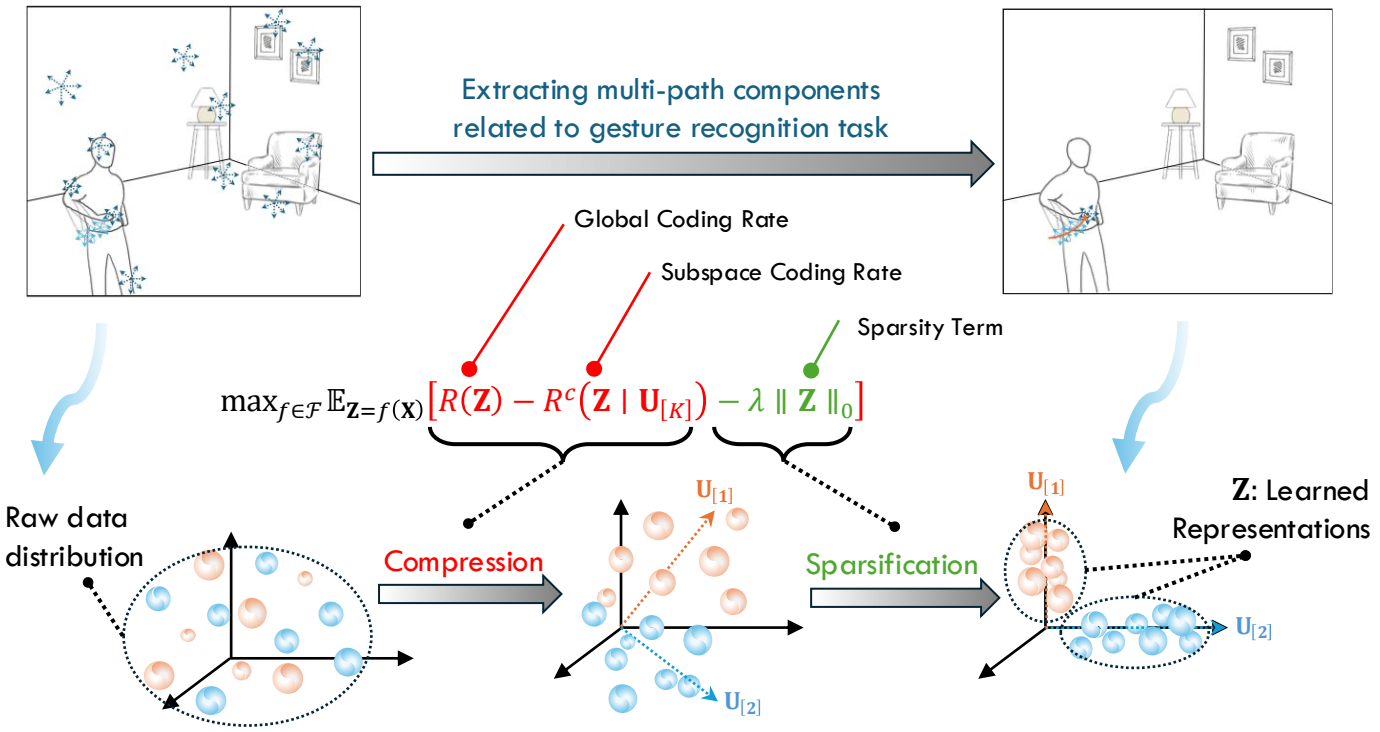
Low-Dimensional Dynamics:
Task-relevant motion is concentrated in a few propagation paths and low-dimensional temporal patterns.

Intrinsic Parsimony of RF Sensing

RF data is high-dimensional in measurement, but low-dimensional and structured in information.

Opportunity: Sparse rate reduction matches this structure

Sparse rate reduction (SRR) extracts low-dimensional, parsimonious structures from high-dimensional measurements, aligning with the objectives of RF sensing.



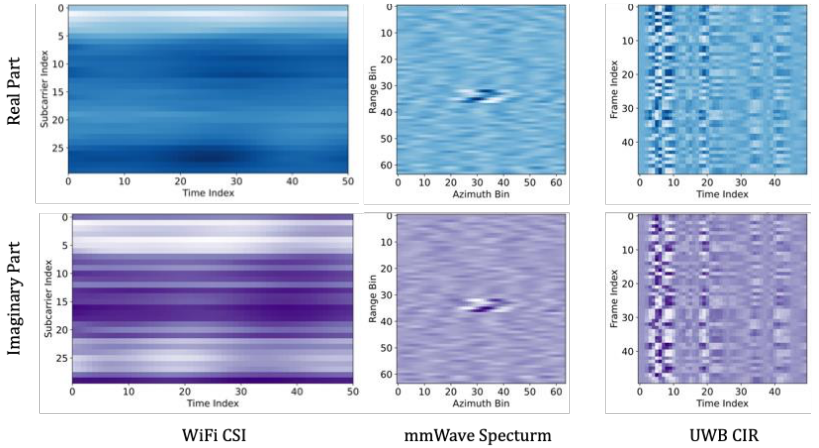
Current Limitation: While SRR-based unrolling yielded the CRATE[1] model for vision, existing frameworks cannot be directly applied to RF sensing.

[1] Yu, Yaodong, et al. "White-box transformers via sparse rate reduction." *Advances in Neural Information Processing Systems* 36 (2023): 9422-9457.

Challenge: Complex-domain Extension

Native Complex-Valued Input:

- RF measurements are inherently complex-valued;
- Discarding or separating phase destroys physical coherence.



Optimization Obstacle: Non-Holomorphic Function

$$\mathcal{L}(Z) = \max_{f \in \mathcal{F}} \mathbb{E}_{\mathbf{Z}=f(\mathbf{X})}\left[R(\mathbf{Z}) - R^c\left(\mathbf{Z} \mid \mathbf{U}_{[K]}\right) - \lambda \parallel \mathbf{Z} \parallel_0\right],$$
where $\mathcal{L}(Z) \in \mathbb{R}$, $Z \in \mathbb{C}^{d \times n}$.

Real-valued outputs from complex inputs make the objective non-holomorphic, rendering standard complex differentiation inapplicable.

Solution: Complex Sparse Rate Reduction with CR-Calculus[2]

Complex Sparse Rate Reduction Objective

$$\mathcal{L}(Z) = \max_{f \in \mathcal{F}} \mathbb{E}_{\mathbf{Z}=f(\mathbf{X})}\left[R(\mathbf{Z}) - R^c\left(\mathbf{Z} \mid \mathbf{U}_{[K]}\right) - \lambda \| \mathbf{Z} \|_0\right],$$

where $\mathcal{L}(Z) \in \mathbb{R}$, $Z \in \mathbb{C}^{d \times n}$, $R^c\left(\mathbf{Z} \mid \mathbf{U}_{[K]}\right) = \frac{1}{2}\sum_{k=1}^{K} \log\det\left(\mathbf{I} + \beta\left(\mathbf{U}_k^H \mathbf{Z}\right)^H \left(\mathbf{U}_k^H \mathbf{Z}\right)\right)$, $R(\mathbf{Z}) = \frac{1}{2}\log\det(\mathbf{I} + \alpha \mathbf{Z}^H \mathbf{Z})$

Interpretation:

- $R(\mathbf{Z})$: preserve global information
- $R^c\left(\mathbf{Z} \mid \mathbf{U}_{[K]}\right)$: compress features within subspaces
- $\| \mathbf{Z} \|_0$: encourage sparse activations

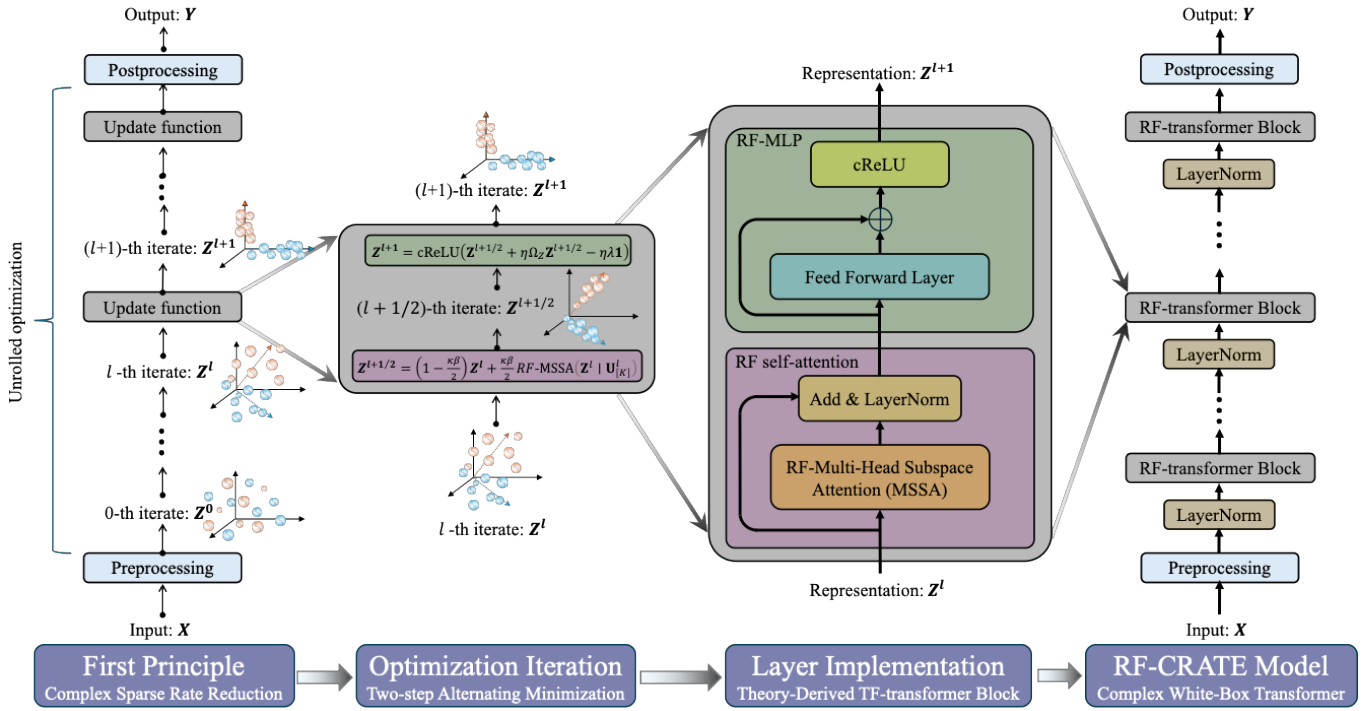
[2] Ken Kreutz-Delgado. 2009. The Complex Gradient Operator and the CR-Calculus. arXiv:0906.4835

RF-CRATE: From an objective to a transformer

Optimizing the Complex Sparse Rate Reduction Objective with Two-step alternating minimization process

$$\mathcal{L}(Z) = \max_{f \in \mathcal{F}} \mathbb{E}_{\mathbf{Z}=f(\mathbf{X})}\left[R(\mathbf{Z}) - R^c\left(\mathbf{Z} \mid \mathbf{U}_{[K]}\right) - \lambda \parallel \mathbf{Z} \parallel_0\right]$$

1 $\min_{f_1 \in \mathcal{F}_1} \mathbb{E}_{\mathbf{Z}^{l+1/2}=f_1(\mathbf{Z}^l)}\left[R^c\left(\mathbf{Z}^{l+1/2} \mid \mathbf{U}^l_{[K]}\right)\right]$ 2 $\min_{f_2 \in \mathcal{F}_2} \mathbb{E}_{\mathbf{Z}^{l+1}=f_2(\mathbf{Z}^{l+1/2})}\left[-R\left(\mathbf{Z}^{l+1}\right) + \lambda \parallel \mathbf{Z}^{l+1} \parallel_0\right]$



½ Compression step: RF self-attention

The first alternating update minimizes the subspace coding rate:

1 $\min_{f_1 \in \mathcal{F}_1} \mathbb{E}_{\mathbf{Z}^{l+1/2}=f_1(\mathbf{Z}^l)}\left[R^c\left(\mathbf{Z}^{l+1/2} \mid \mathbf{U}_{[K]}^l\right)\right]$

Using a CR-Calculus gradient step and a first-order Neumann approximation:

$$\boldsymbol{Z}^{l+1/2} = \left(1 - \frac{\kappa\beta}{2}\right)\boldsymbol{Z}^l + \frac{\kappa\beta}{2}\mathrm{RF{-}MSSA}\left(\mathbf{Z}^l \mid \mathbf{U}_{[K]}^l\right)$$

Meaning:

- **Subspace Projection:** Projects token embeddings into learned subspaces corresponding to distinct task-related patterns.
- **Pattern Association:** Groups and correlates related tokens within these specialized subspaces.

½ Compression step: RF self-attention

The first alternating update minimizes the subspace coding rate:

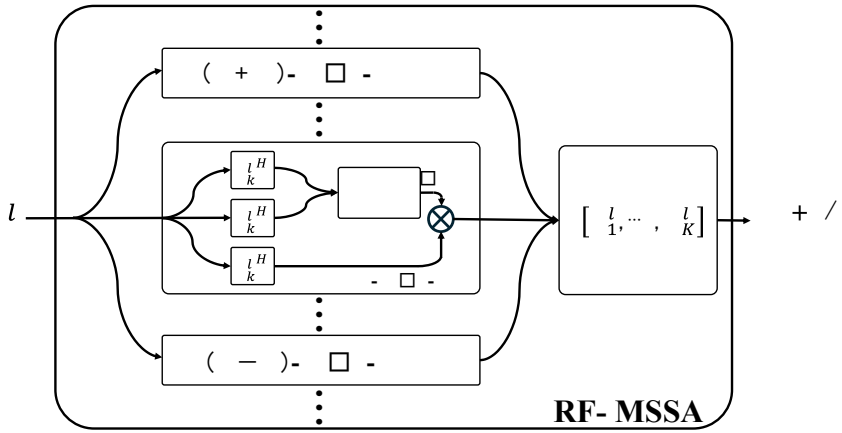
1 $\min_{f_1 \in \mathcal{F}_1} \mathbb{E}_{\mathbf{Z}^{l+1/2}=f_1(\mathbf{Z}^l)}\left[R^c\left(\mathbf{Z}^{l+1/2} \mid \mathbf{U}_{[K]}^l\right)\right]$

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$$\mathrm{RF{-}MSSA}\left(\mathbf{Z}^l \mid \mathbf{U}_{[K]}^l\right) = \beta\left[\mathbf{U}_1^l, \cdots, \mathbf{U}_K^l\right]\begin{bmatrix}\mathrm{RF{-}SSA}\left(\mathbf{Z}^l \mid \mathbf{U}_1^l\right)\\ \vdots \\ \mathrm{RF{-}SSA}\left(\mathbf{Z}^l \mid \mathbf{U}_K^l\right)\end{bmatrix}$$

$$\mathrm{RF{-}SSA}\left(\mathbf{Z}^l \mid \mathbf{U}_k^l\right) = \underbrace{\mathbf{U}_k^{l\,H}\mathbf{Z}^l}_{\text{Value}}\,\mathcal{S}\Big(\underbrace{\left(\mathbf{U}_k^{l\,H}\mathbf{Z}^l\right)^H}_{\text{Query}}\,\underbrace{\left(\mathbf{U}_k^{l\,H}\mathbf{Z}^l\right)}_{\text{Key}}\Big)$$



⅔ Sparsification step: from LASSO to RF-MLP

The second update begins with:

2 $\min_{f_2 \in \mathcal{F}_2} \mathbb{E}_{\mathbf{Z}^{l+1}=f_2(\mathbf{Z}^{l+1/2})}\left[-R\left(\mathbf{Z}^{l+1}\right) + \lambda \| \mathbf{Z}^{l+1} \|_0\right]$

For a tractable update, the derivation:

1. Relaxes (l_0 norm) to (l_1 norm);
2. Adds an elementwise non-negativity constraint;
3. Upper-bounds the smooth term ($-R\left(\mathbf{Z}^{l+1}\right)$) with CR-Calculus;
4. Applies a proximal sparsification step;

The resulting operation is a **complex residual MLP:**

$$Z^{l+1} = \mathrm{cReLU}\left(\left(1 + \frac{16}{9(1+\alpha)}\Omega_Z\right) Z^{l+1/2} - \frac{16\lambda}{9\alpha}\mathbf{1}\right)$$

½ Sparsification step: from LASSO to RF-MLP

The second update begins with:

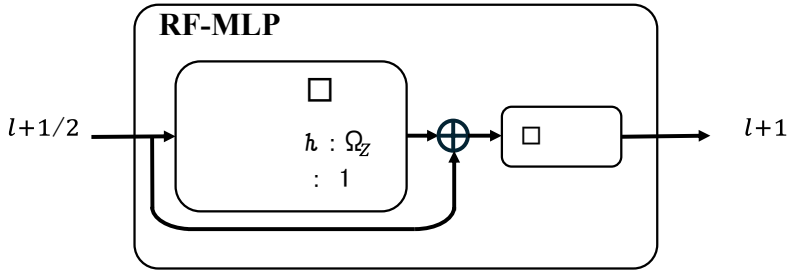
2 $\min_{f_2 \in \mathcal{F}_2} \mathbb{E}_{\mathbf{Z}^{l+1}=f_2(\mathbf{Z}^{l+1/2})}\left[-R\left(\mathbf{Z}^{l+1}\right) + \lambda \| \mathbf{Z}^{l+1} \|_0\right]$

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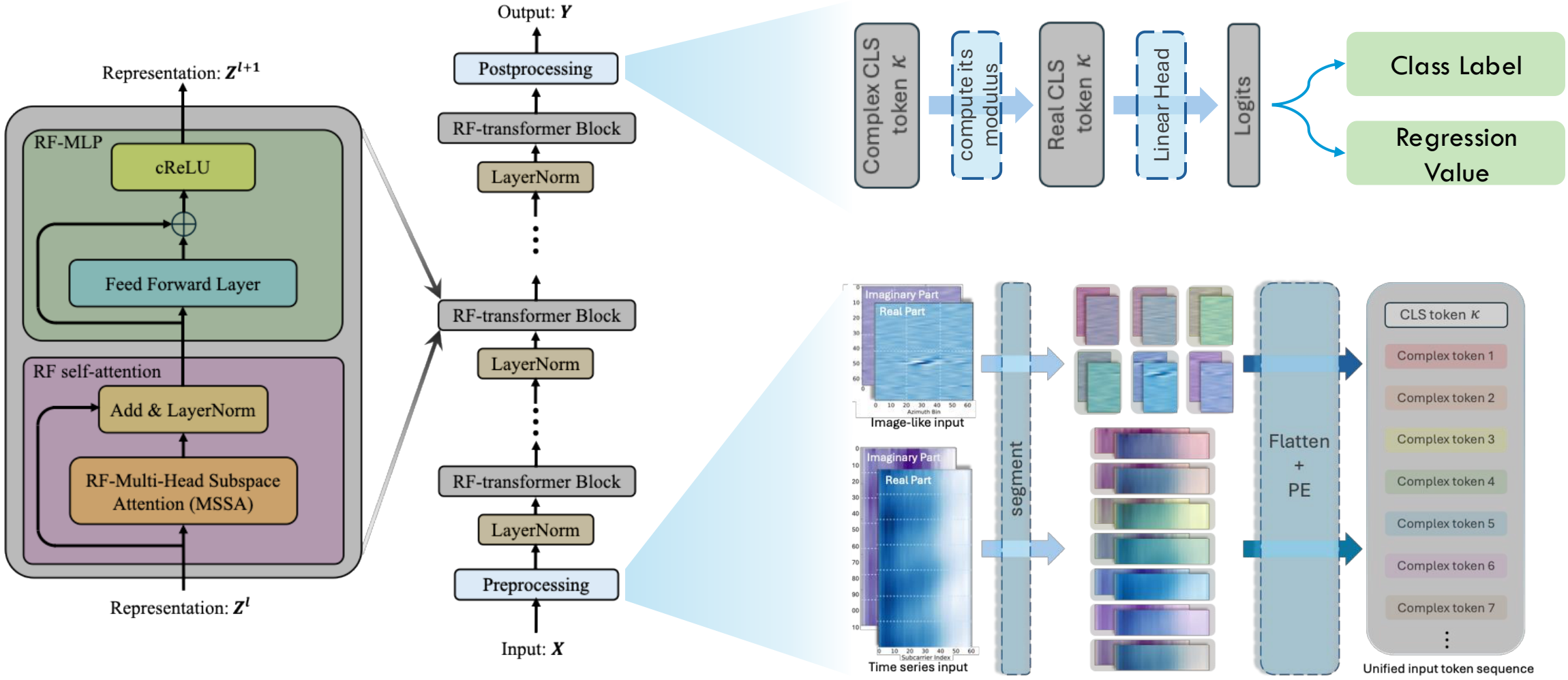
$$Z^{l+1} = \mathrm{cReL\,U}\left(\left(1 + \frac{16}{9(1+\alpha)}\Omega_Z\right) Z^{l+1/2} - \frac{16\lambda}{9\alpha}\mathbf{1}\right)$$

Relax metric tensor Ω_Z into **unconstrained learnable weights.**

$$RF - MLP\left(\mathbf{Z}^{l+1/2}\right) \triangleq \mathrm{cReLU}\left(\mathbf{Z}^{l+1/2} + \eta\Omega_Z \mathbf{Z}^{l+1/2} - \eta\lambda\mathbf{1}\right)$$



Complete RF-CRATE pipeline



What the model reveals after training

The derived objective predicts two observable properties:

1. **Low inter-subspace correlation:** Different subspaces should capture complementary factors.
2. **Sparse feature activation:** Only a few dimensions should carry large magnitude.

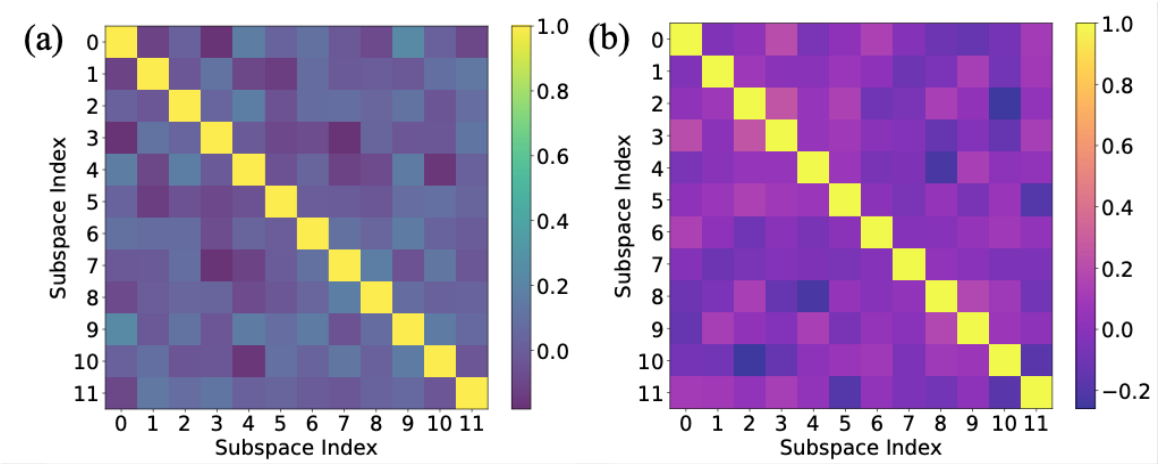


Fig. 6: “low off-diagonal correlation”

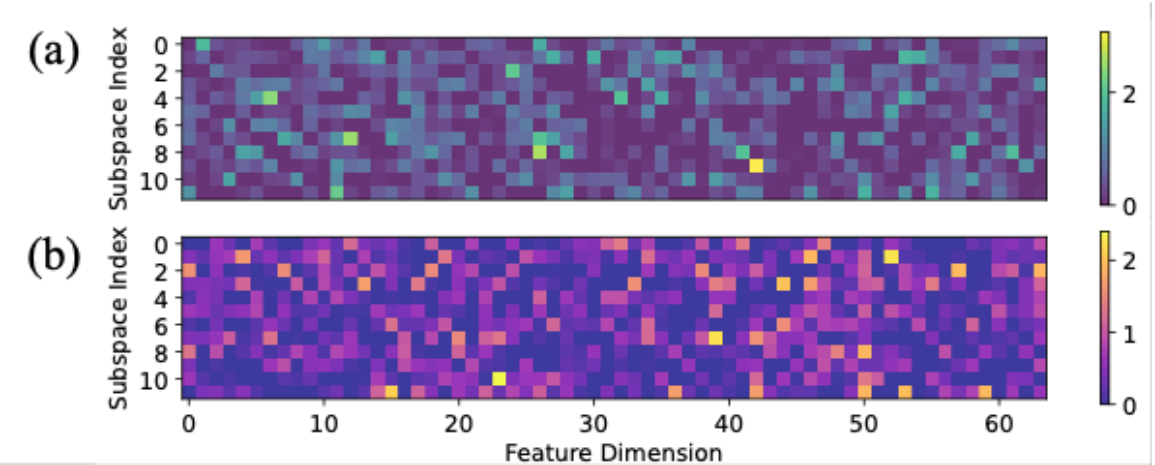


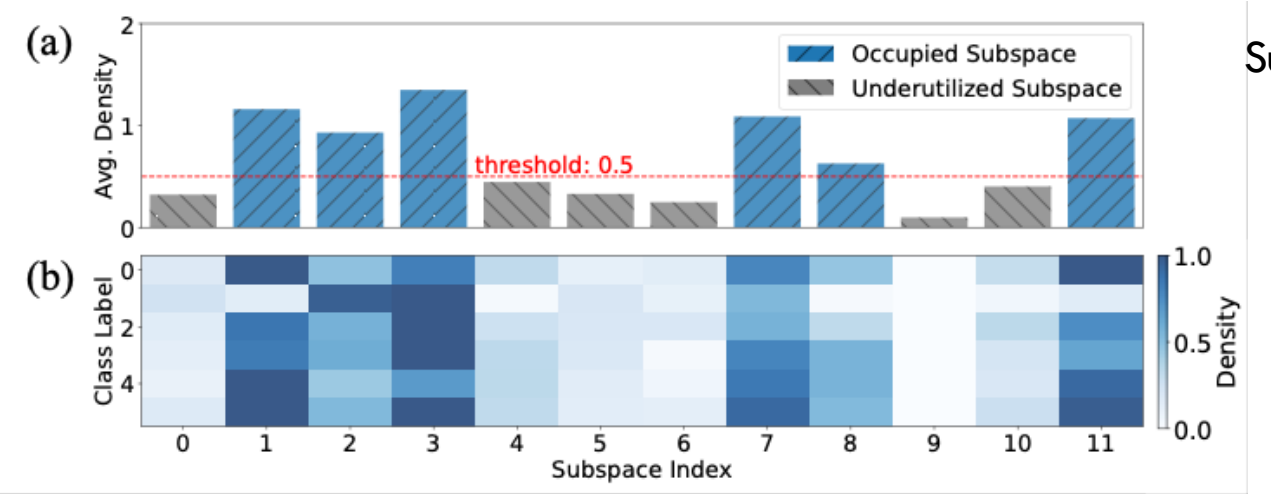
Fig. 7: “few active dimensions”

Empirical inspection supports both behaviors:

- Correlation concentrates near the diagonal.
- Most feature dimensions remain close to zero.

A problem exposed by the white-box structure

“Subspace collapse”: the representation is structured, but its capacity is imbalanced.



Subspace occupancy analysis shows:

- Different classes activate different density patterns.
- A small subset of subspaces can dominate.
- Underused subspaces represent wasted capacity.

“Subspace regularization”: Prevents feature energy from concentrating in a few dominant subspaces, optimizing capacity usage and training efficiency.

For subspace (k), define batch (N) density:

$$\rho_k = \frac{1}{N}\sum_{i=1}^{N}\left|Z_{k,i}\right|$$

$$SSR = \sum_{k=1}^{K}(\rho_k - \rho_{min})$$

$$\rho_{min} = \min_{k\in[K]} \rho_k$$

Evaluation scope, Baselines, and experimental protocol

3 RF modalities

4 Datasets

6 Sensing tasks

WiFi CSI

- Widar3.0: gesture recognition
- GaitID: gait identification
- WiP-pose: 3D pose estimation
- WiP-breath: respiratory-rate estimation

FMCW mmWave

- HuPR: 2D pose estimation

IR-UWB

- OPERAnet: activity recognition

Metrics: **accuracy ↑ · MPJPE ↓ · BPME ↓**

7 Baselines

Generic transformer baselines:

- Swin Transformer
- Swin Transformer V2
- CRATE

DWS-specialized baselines

- RF-Net
- STFNets
- SLNet
- Widar3.0

Training

- AdamW, initial learning rate (5×10^{-5})
- Cosine learning-rate schedule
- Up to 100 epochs
- Early stopping with patience 10
- Matched preprocessing, splits, and optimization where possible

Overall result

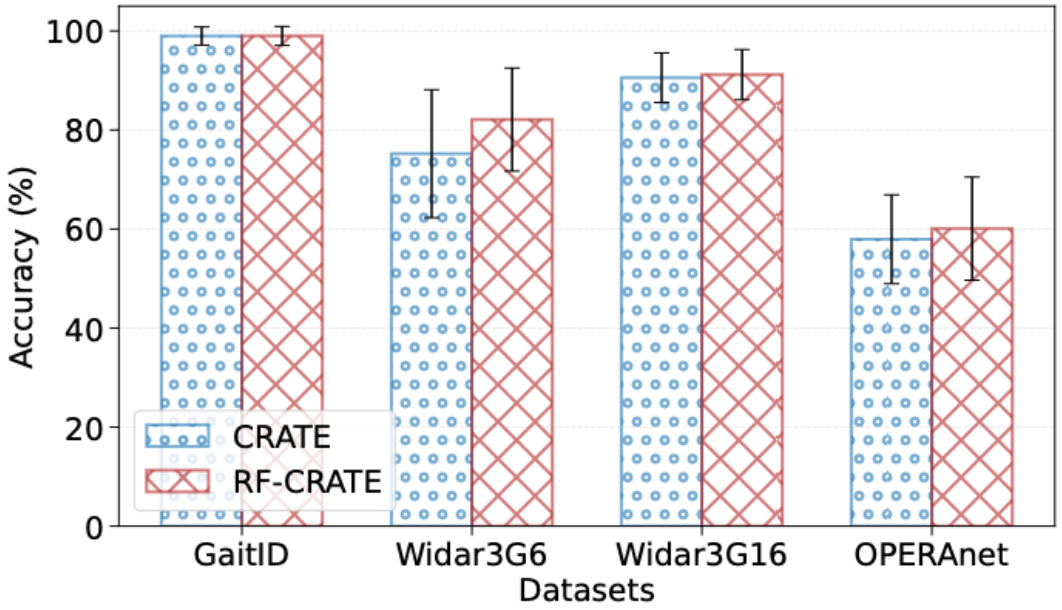
Model Type	Models	WiFi Gesture (Acc. ↑)	WiFi Gait (Acc. ↑)	WiFi 3D Pose (MPJPE ↓)	WiFi Breath (BPME ↓)	mmWave Pose (MPJPE ↓)	UWB Activity (Acc. ↑)
Transformer	Swin-T	84.25±9.96	96.86±3.11	28.70±26.23	2.44±0.86	10.61±6.02	67.02±9.16
	Swin-T V2	55.65±18.45	92.70±4.90	28.90±25.99	2.44±0.77	10.40±6.54	62.61±6.83
	CRATE	75.20±12.92	98.96±1.83	28.66±26.28	2.44±0.85	24.93±11.96	57.92±8.97
DWS	RF-Net	55.33±15.91	99.00±1.88	N/A	3.57±0.98	8.37±5.91	57.87±5.28
	STFNets	50.61±17.48	95.30±4.14	N/A	N/A	47.20±21.44	58.59±7.59
	SLNet	73.67±13.11	94.79±4.14	N/A	N/A	N/A	N/A
	Widar3.0	50.45±18.29	82.01±6.53	N/A	N/A	62.40±15.70	N/A
	RF-CRATE	82.10±10.41	98.98±1.91	28.12±27.02	2.44±0.83	17.09±9.13	60.07±10.41

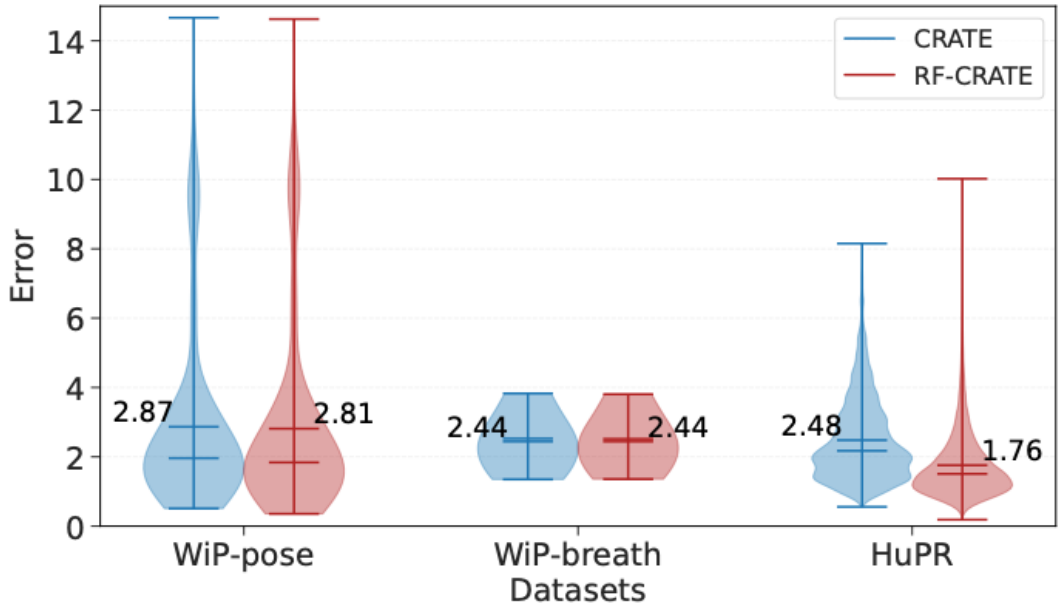
❖ Underline indicates the best DWS-specialized baseline

RF-CRATE achieves competitive performance across diverse sensing tasks while maintaining a fully transparent, white-box design.

Does the complex extension help?

RF-CRATE versus CRATE





+3.39%

Classification Accuracy Gain

10.34%

Regression error reduction

Complex numbers alone are not enough

“complex ≠ automatically better”

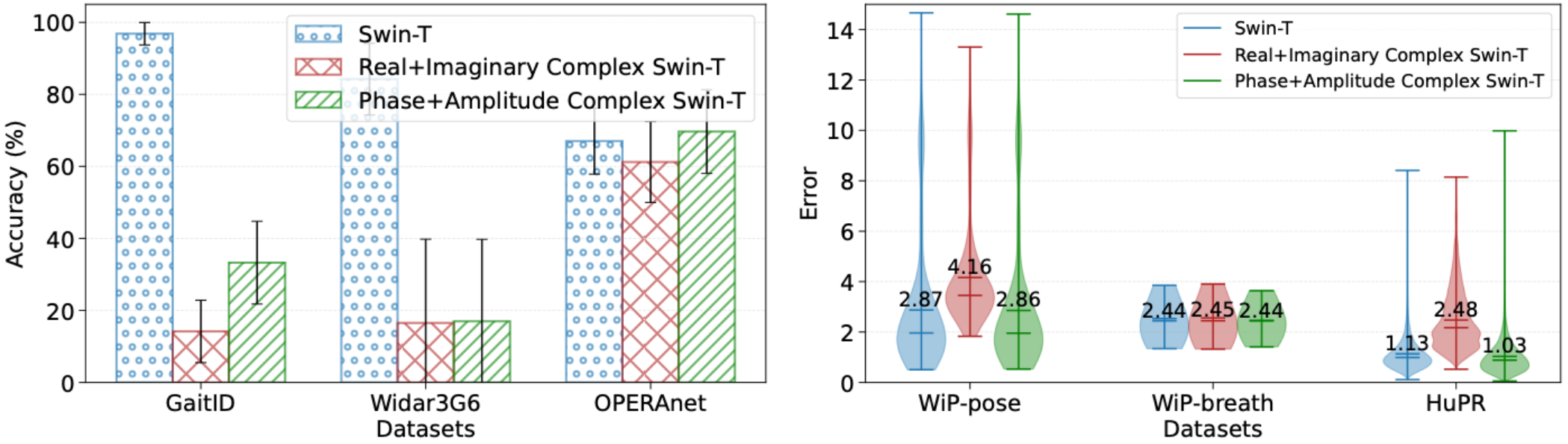
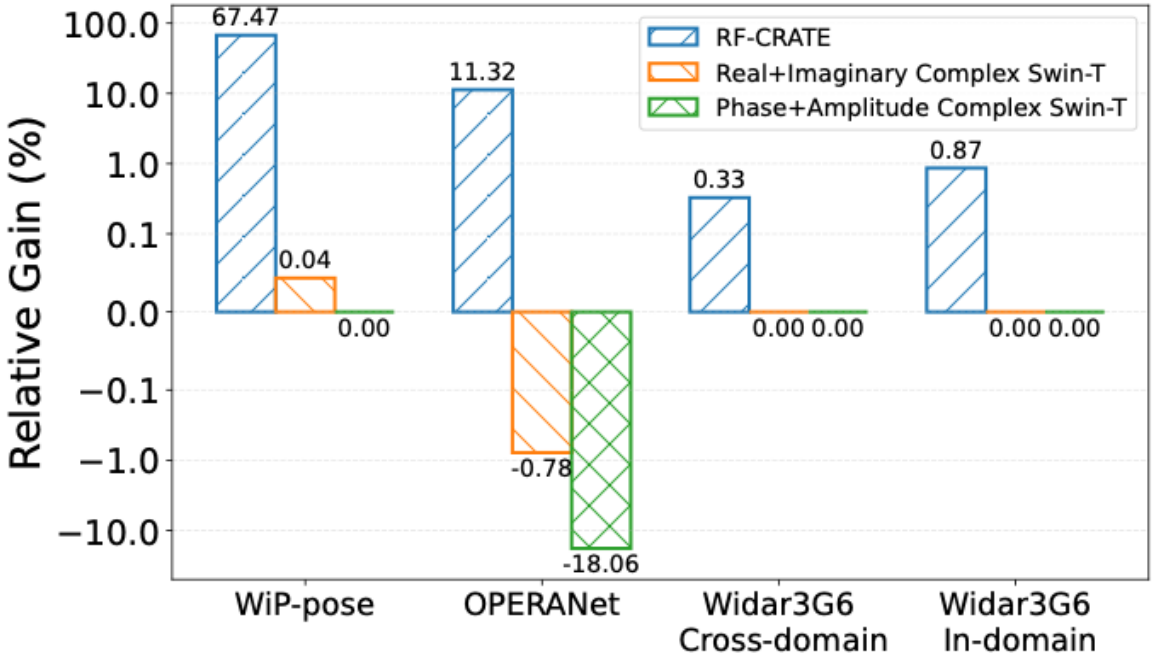


Fig. 8: “Swin-T complex-domain ablation”

Conclusion: the gain is not simply “use complex numbers.”

The complex representation must align with the architecture and learning objective.

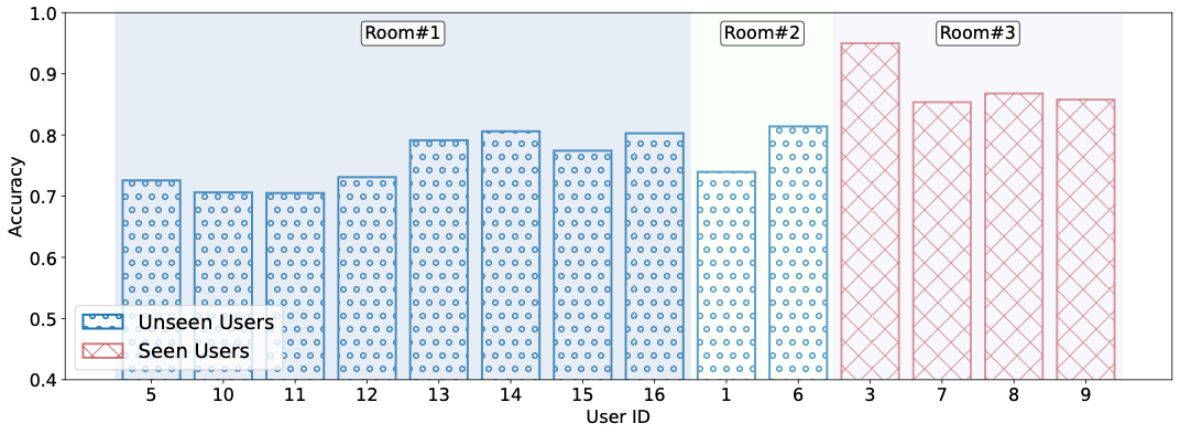
SSR works when the architecture has real subspaces



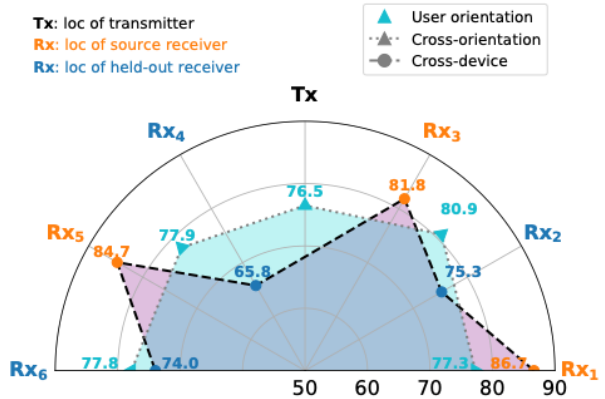
- Adding SSR to RF-CRATE yields:
  19.98% average improvement across four evaluated settings
- Adding the same loss to complex Swin-T is neutral or harmful.

White-box structure provides a principled “hook” for intervention.

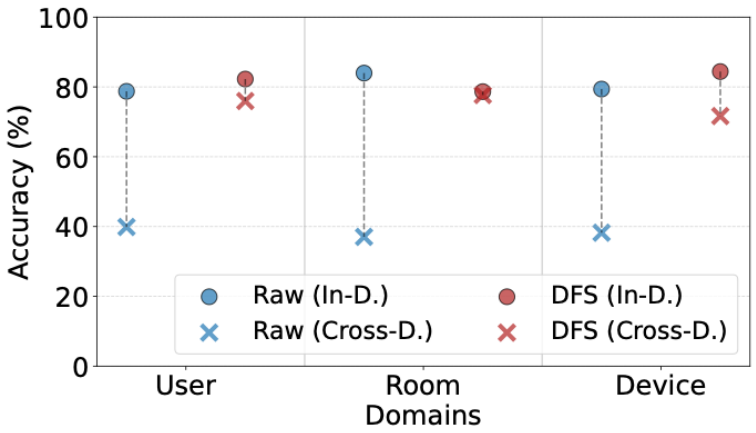
Robustness and practical ablations



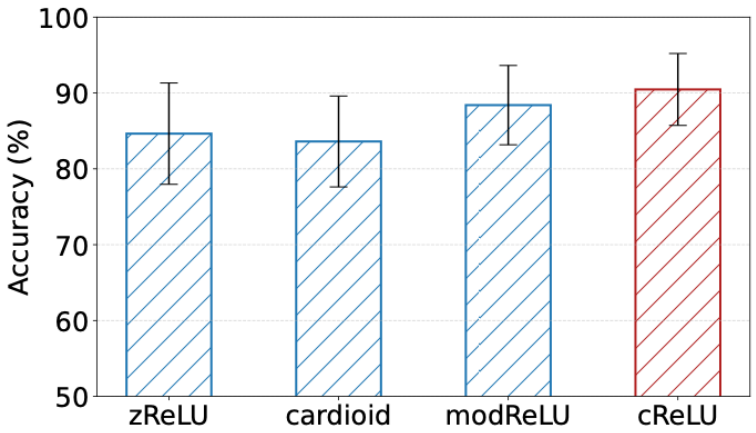
Strong performance across seen and unseen users



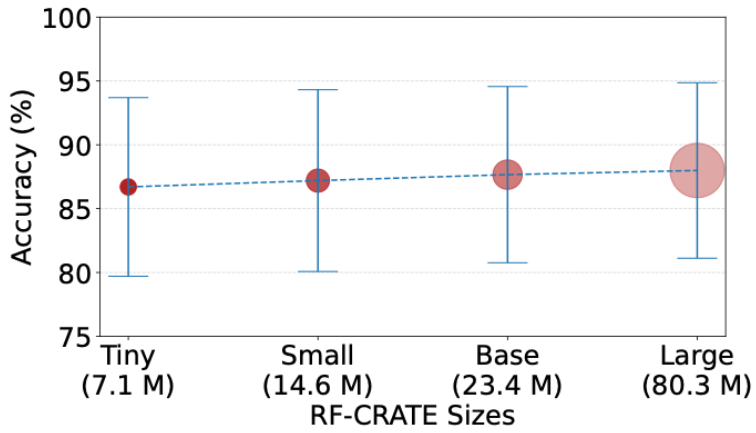
Stable behavior across held-out receivers and user orientations



DFS spectra generalize better than raw CSI under domain shift.

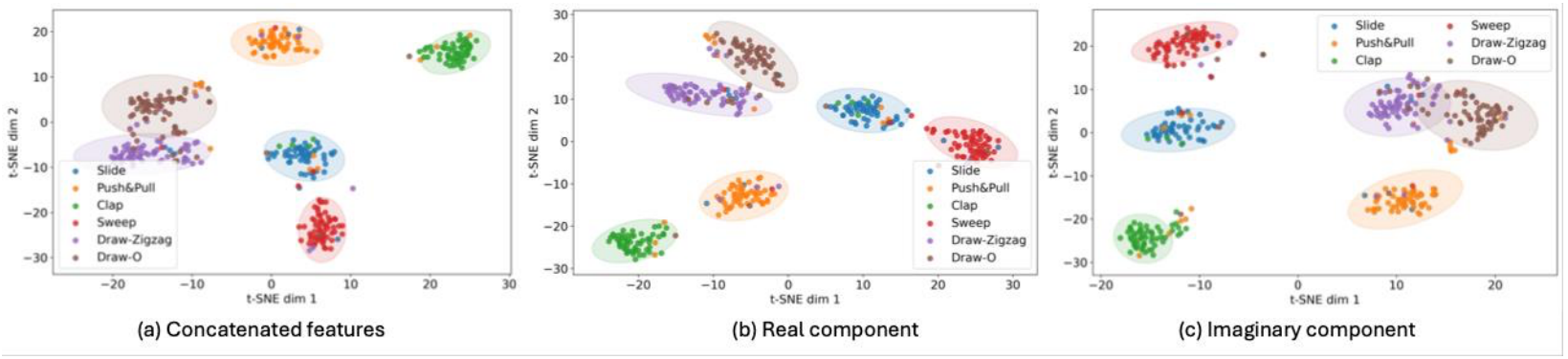


Derived cReLU performs best: 90.47% on Widar3G6.



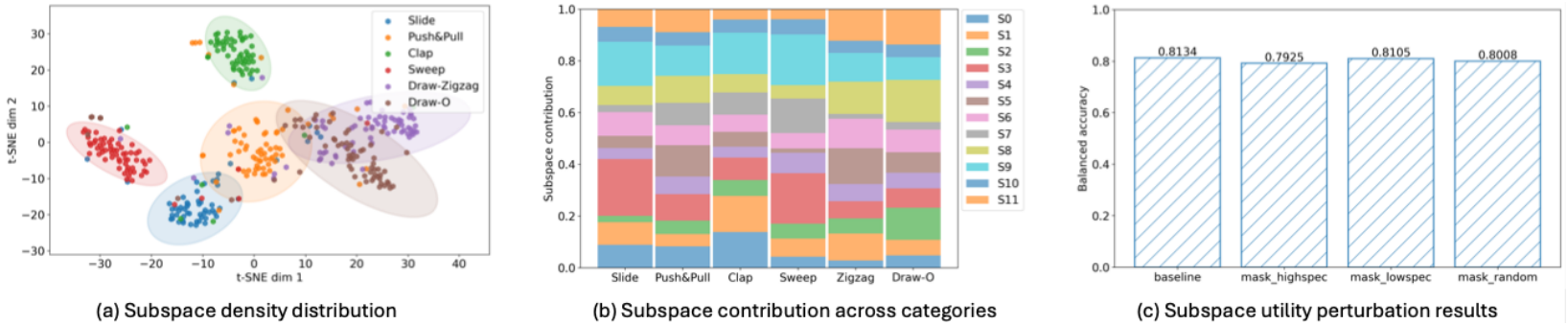
RF-CRATE performance w.r.t. different model sizes.

What do the learned representations mean?



(a) Concatenated features (b) Real component (c) Imaginary component

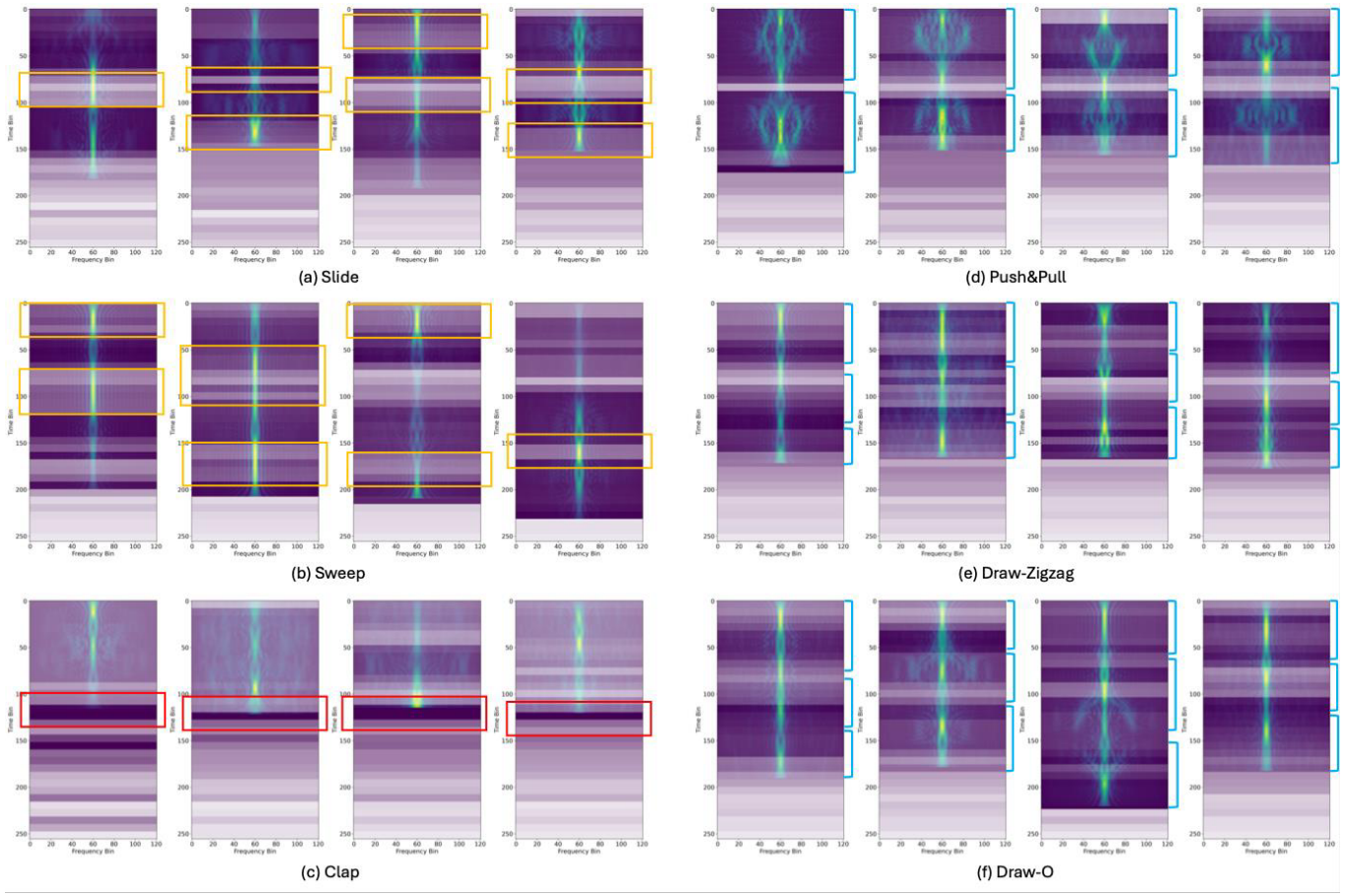
- Final complex features form gesture-dependent clusters.
- Real and imaginary components are each discriminative.



(a) Subspace density distribution (b) Subspace contribution across categories (c) Subspace utility perturbation results

- Subspace-density vectors preserve task semantics and intra-class variation.
- Multiple subspaces contribute jointly rather than acting as isolated class detectors.

Attention can align with physics—and expose shortcuts



Attention over the DFS spectrum shows plausible physical alignment:

- Static energy near **0 Hz** is suppressed.
- **Push & Pull** produces a two-stage attention pattern.
- **Draw-Zigzag** and **Draw-O** share similar three-stage signatures.

But the model also finds a shortcut:

- For **Clap,** attention focuses on the transition to zero-padding.

Takeaways, limitations, and outlook

Three takeaways

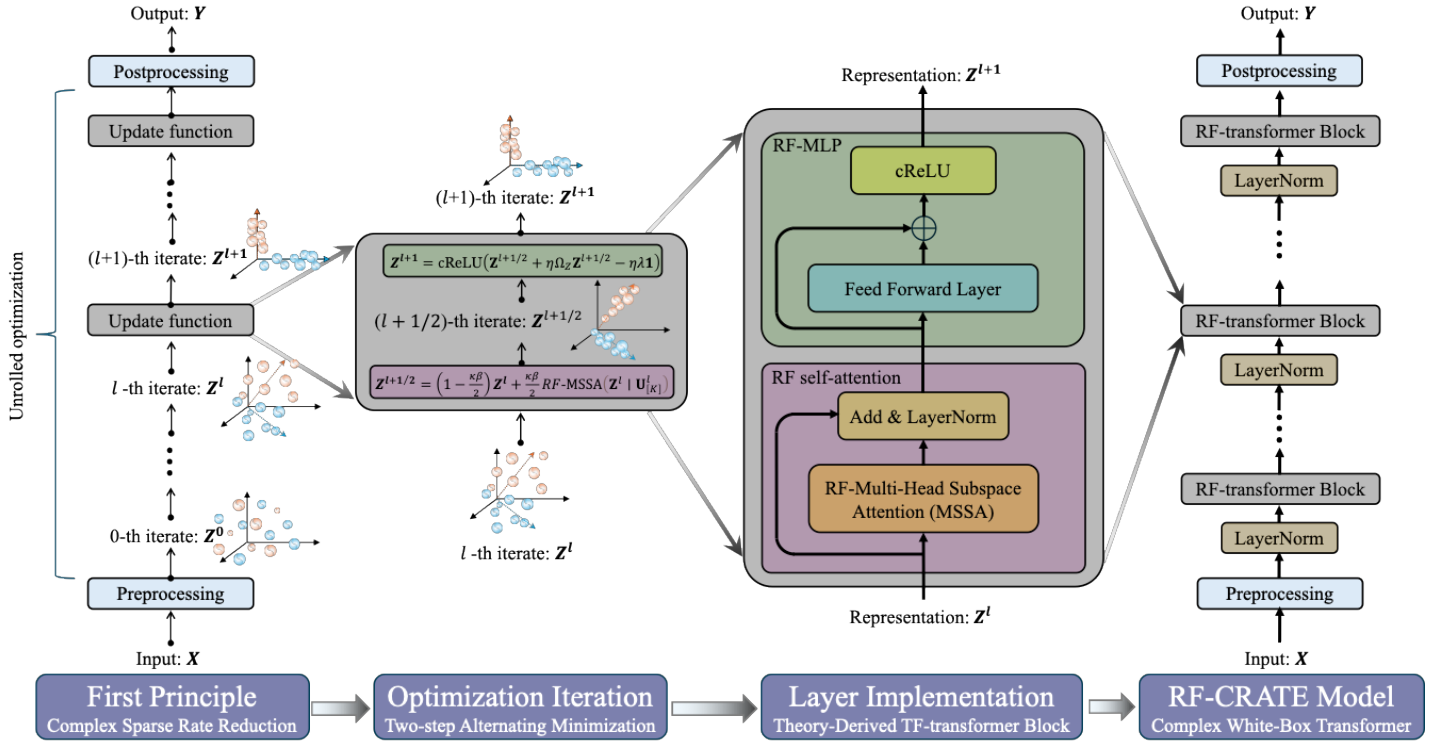
1. A deep RF sensing architecture can be derived from a complex sparse rate reduction objective.
2. Complex-domain modeling helps when it is aligned with the architecture — not by itself.
3. White-box structure enables targeted analysis and regularization such as SSR.

Current limitations

- Mathematical transparency is not yet human-semantic interpretability and physical interpretable.
- Edge deployment remains difficult because attention cost grows with sequence length.
- The derivation still uses practical relaxations and learnable approximations.

Outlook

Move from mathematically grounded models toward RF systems that are also physically grounded, diagnosable, and deployable.



Thank you

Code & Data Available: RF-CRATE is fully open-sourced at https://github.com/aiot-lab/RF-CRATE

TAP3D: Thermal-Assisted 3D Human Point Clouds

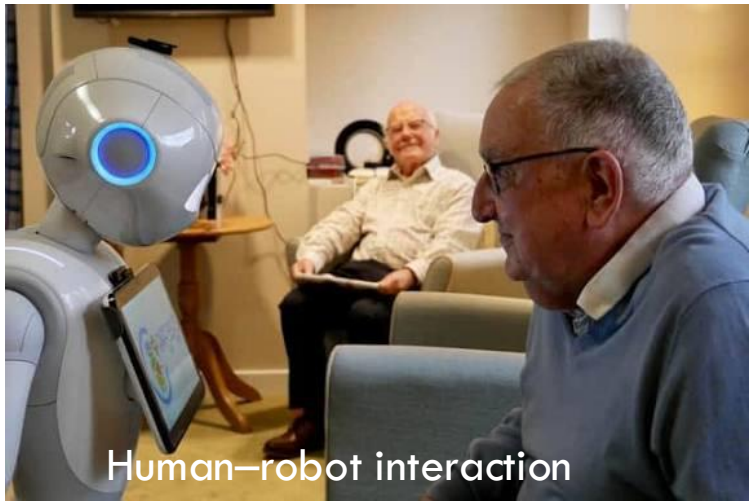
Xie ZHANG, Chengxiao Li, Xuan Liu, Chenshu Wu

The University of Hong Kong



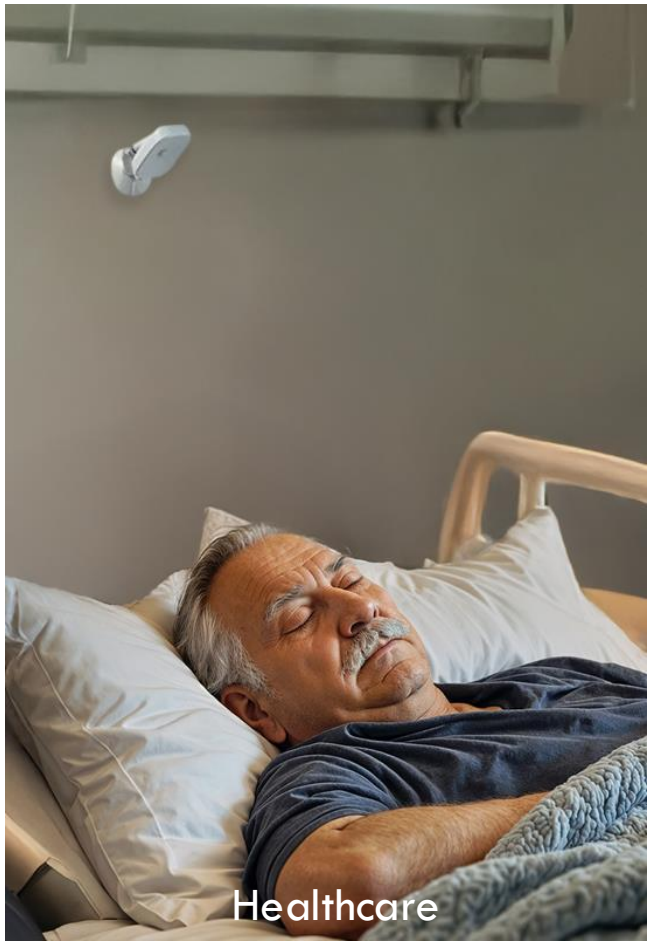
MobiCom'26

Non-Contact Human Sensing Techniques



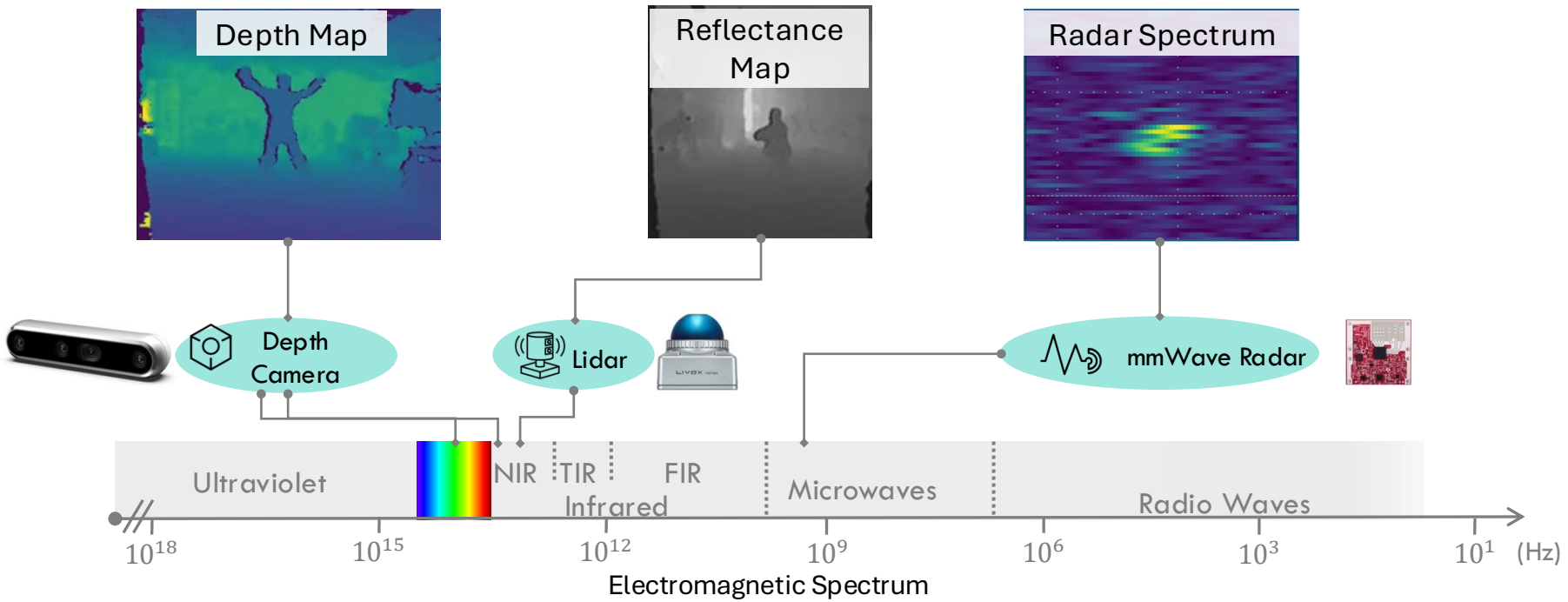






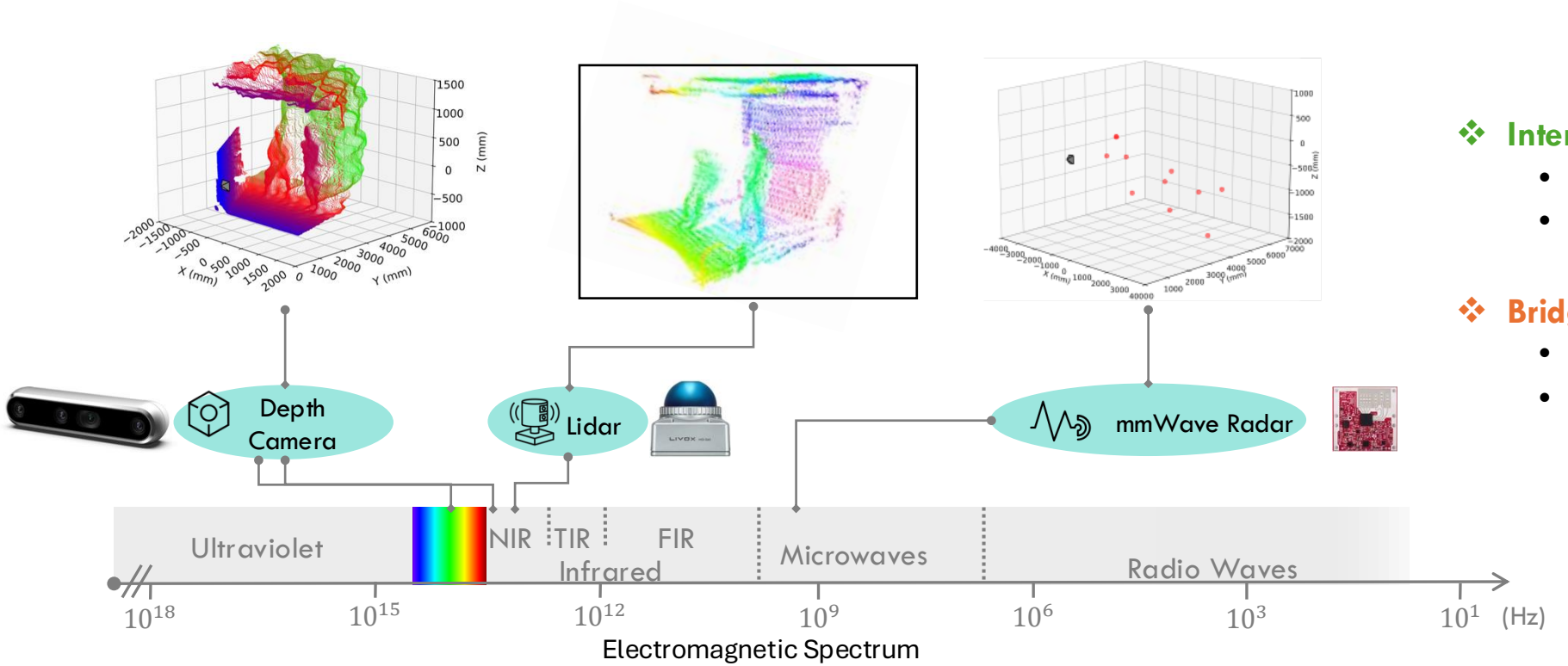
Enabling Human-Centric Applications through
Spatial and Behavioral Understanding

Distinct Signal Representations from 3D Sensing Devices



3D Body Point clouds as Intermediate/Bridge Representation

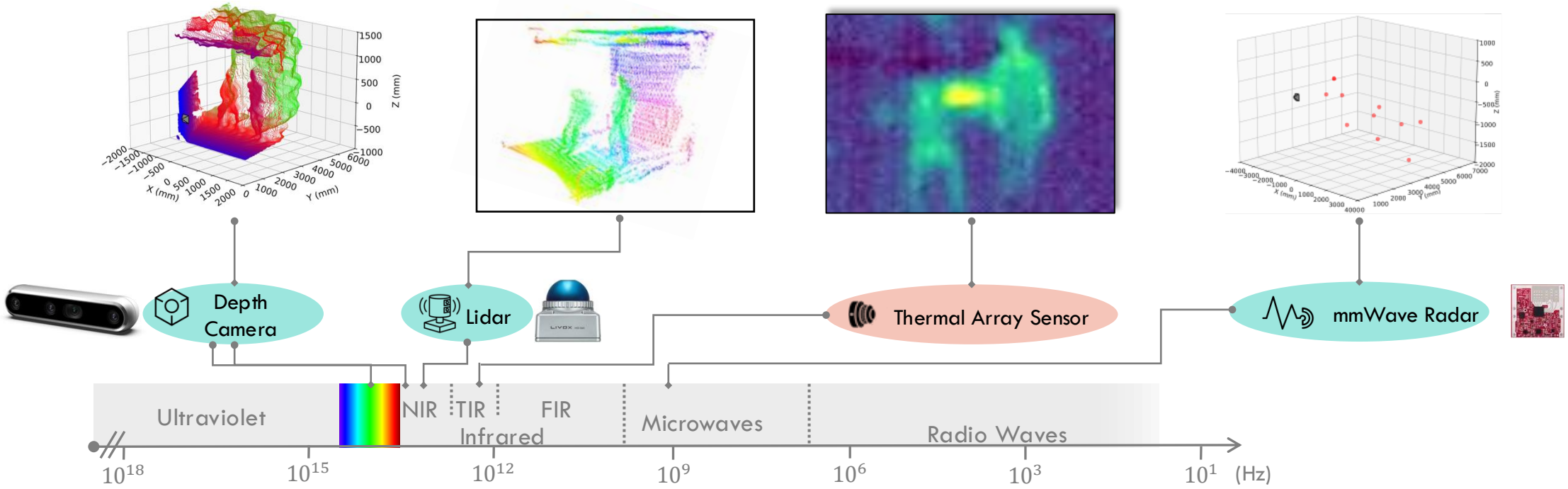
Modality	Cost Resolution Balance	Privacy Preserving	Human Sensitivity	Smoke/Fog Robust	No Emission Risk	Light Immunity	No Cross-device interference
Depth Camera			✅		✅		✅
LiDAR		✅					
Radar		✅		✅		✅	



- ❖ **Intermediate representation**:
  - Detailed pose & shape
  - Diverse downstream tasks
- ❖ **Bridge representation**:
  - Cross-modal alignment
  - Multi-sensor fusion

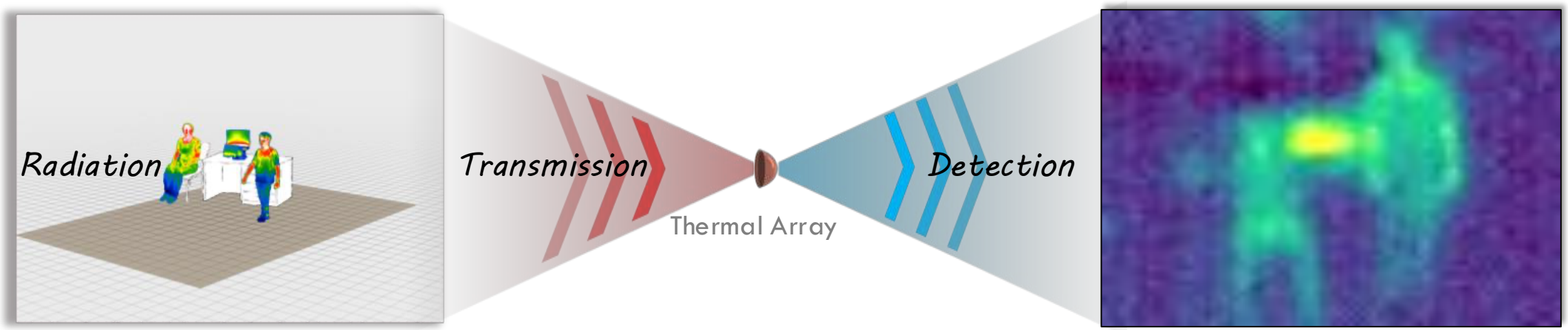
3D Body Point clouds as Intermediate/Bridge Representation

Modality	Cost Resolution Balance	Privacy Preserving	Human Sensitivity	Smoke/Fog Robust	No Emission Risk	Light Immunity	No Cross-device interference
Depth Camera			✅		✅		✅
LiDAR		✅					
Radar		✅		✅		✅	
Thermal Array	✅	✅	✅	✅	✅	✅	✅



Electromagnetic Spectrum

Thermal Array-based Sensing: Principles and Properties



Fully-Passive Sensing

- Illumination-invariant
- No cross-device interference
- No active-emission safety concerns

Long-Wavelength (8–15 μm)

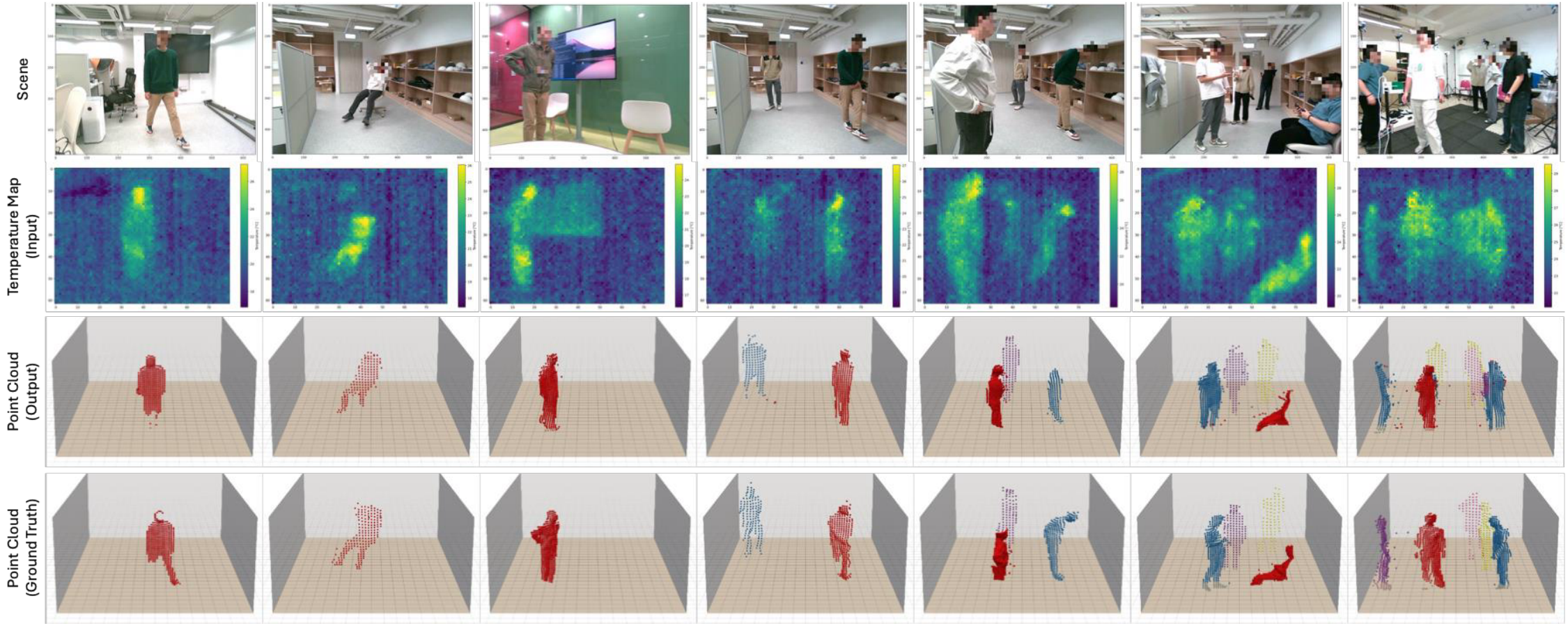
- Privacy-preserving body heat signatures
- Robust to glare, smoke, and fog
- Sensitive to human body

Multi-Physics Encoding

- Preserves 2D spatial heat patterns
- Encodes temperature, emissivity, and range-related cues

Can we enable 3D human point cloud generation using thermal array sensor?

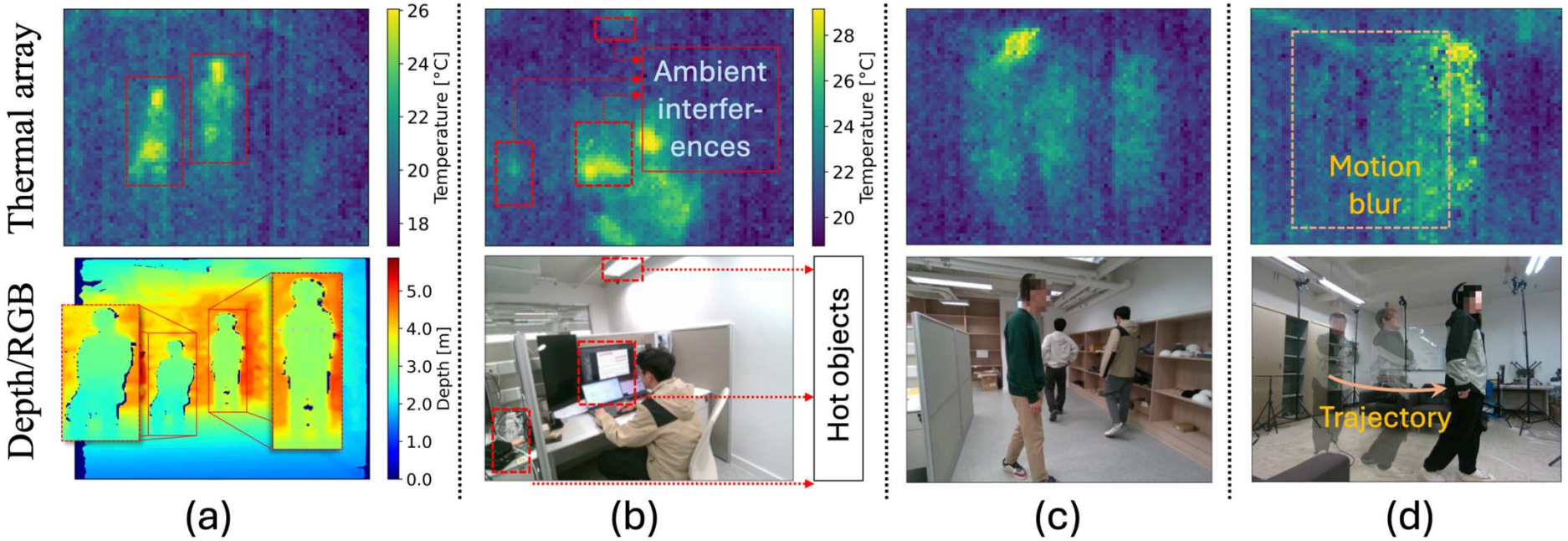
Visualization of TAP3D Output



Results of TAP3D: Thermal-Assisted 3D Human Point Clouds

Challenges in TAP3D

Challenges in generating point clouds from thermal data:



(a) Depth estimation: thermal intensity is a poor proxy for geometry.
(b) Thermal interference: ambient heat sources can cause false positives.
(c-d) Multi-user ambiguity: adjacent signatures may merge and motion blur further obscures body boundaries.

Challenges in TAP3D

Pilot study

Model:

- U-Net learning model.
- Input: temperature map.
- Output: depth map & User index map.

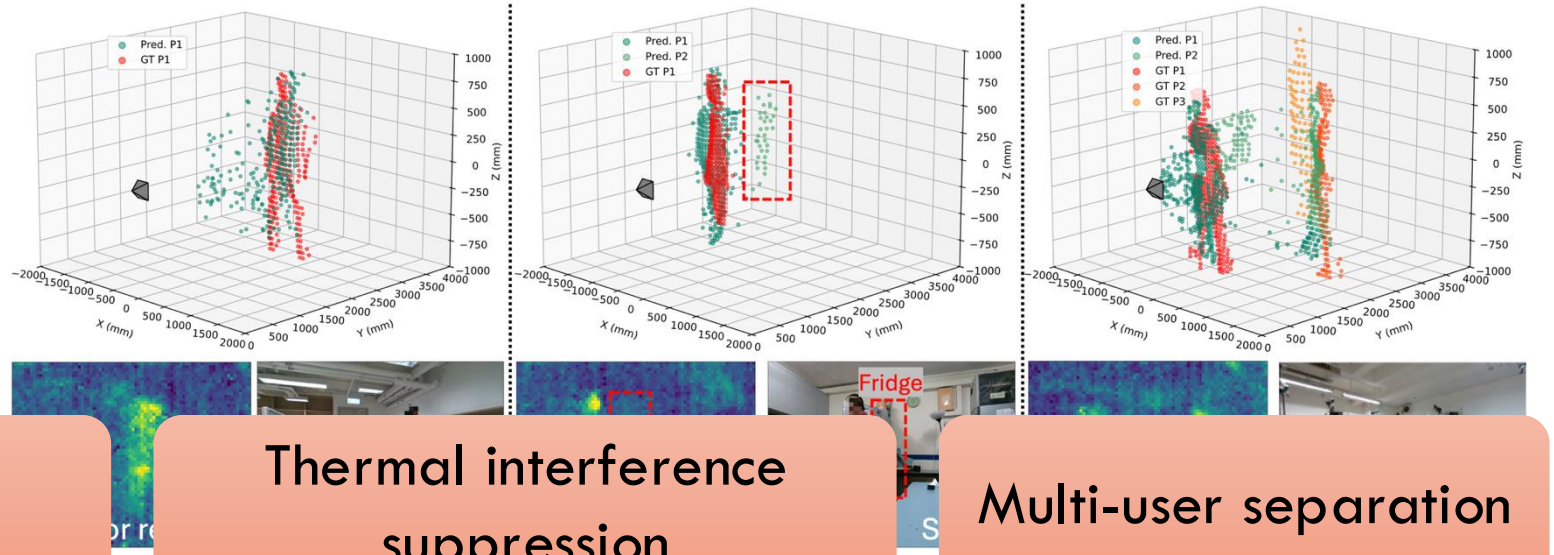
Dataset:

- Training set: 17,956 sample.
- Test
- 5 vo
  envi

results:

Scenario	DCD (cm) ↓	F1 Score ↑	MAE (cm) ↓
Overall	22.6	0.88	31.94
Single-User	18.5	0.91	26.83
Multi-User	34.2	0.82	51.72

SRCD: square root of Chamfer Distance
MAE: mean absolute error of depth estimation



Accurate depth recovery

Thermal interference suppression

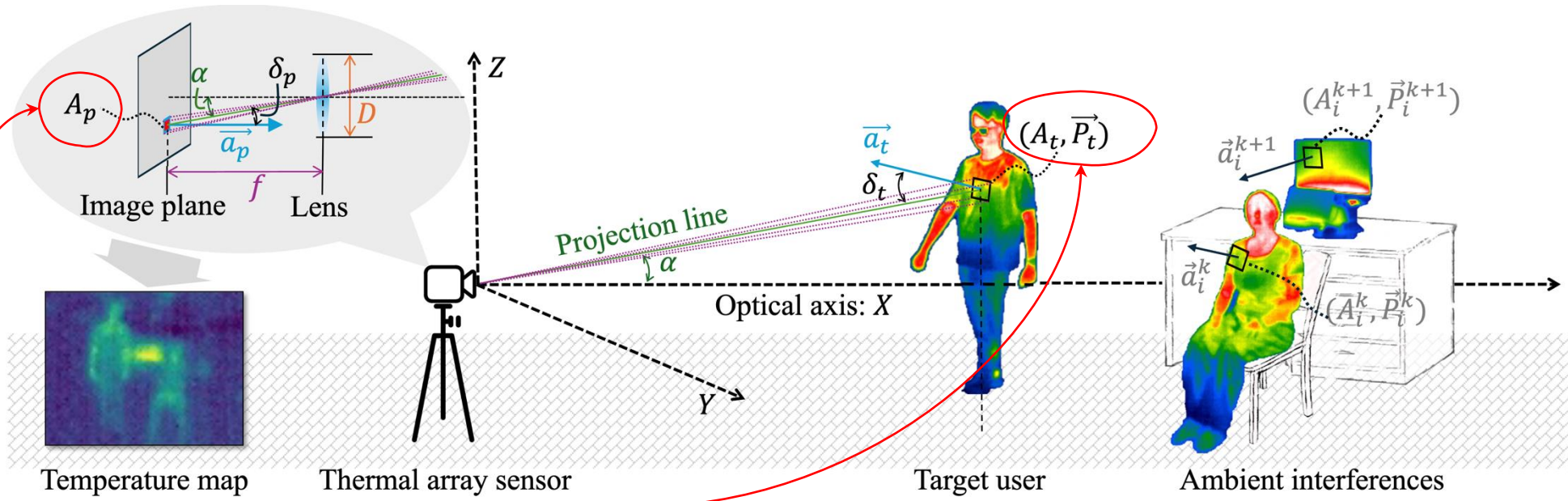
Multi-user separation

Qualitative results from the pilot study illustrating the key challenges:

- Left: Depth estimation error, where the predicted (Pred.) point cloud deviates from the ground truth (GT).
- Middle: Thermal interference, where a warm refrigerator (dotted red block) confuses the detection.
- Right: Multi-user separation failure, where two individuals are incorrectly merged into one point cloud.

Physical Model

- Accurate depth recovery
- Thermal interference suppression
- Multi-user separation



$$\hat{T}_p\left(\overrightarrow{P_t}; T_t, \epsilon_t\right) = \frac{C_2}{C_3 \ln\left(\frac{C_1}{V_1+V_2}+1\right)} + \frac{C_4}{C_3} \quad (1)$$

$$V_2 = \frac{GD^2 A_p \xi_t \zeta \left\langle \overrightarrow{P_t}, \vec{x} \right\rangle^7 e^{-\gamma|\overrightarrow{P_t}|}}{4f^4 \, |\overrightarrow{P_t}|^5} (1-\epsilon_t) \sum_{k=1}^{K} e^{-\gamma|\overrightarrow{P_t}-\overrightarrow{P_i^k}|} S_{\lambda,k} \quad (3)$$

$$V_1 = \frac{GD^2 \zeta A_p}{4f^2} \frac{\left\langle \overrightarrow{P_t}, \vec{x} \right\rangle^4 e^{-\gamma|\overrightarrow{P_t}|}}{|\overrightarrow{P_t}|^4} \epsilon_t \int_{\lambda_1}^{\lambda_2} M_\lambda^{\mathrm{BB}}(T_t) d_\lambda \quad (2)$$

TAP3D Model Design

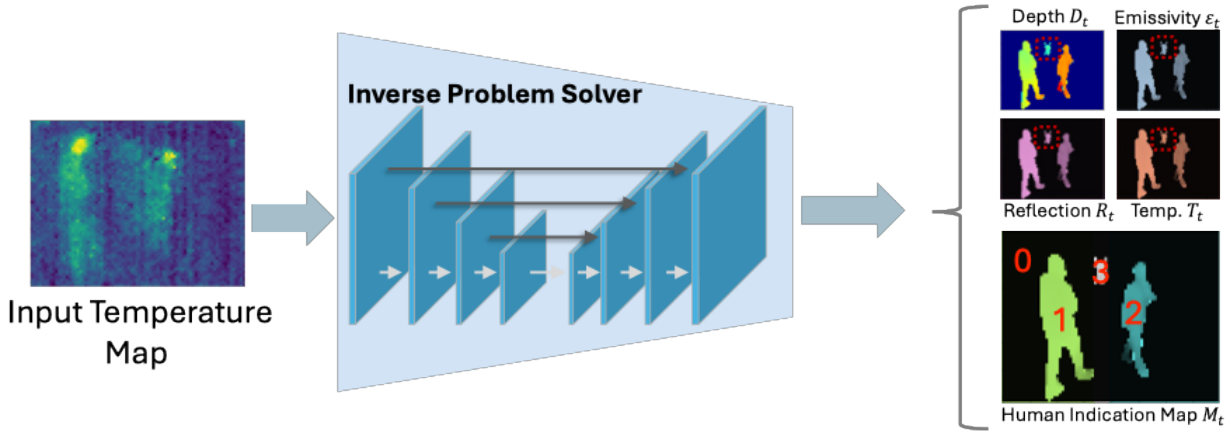
Accurate depth recovery

Thermal interference suppression

Multi-user separation

Root cause of depth estimation difficulty:

Depth nonlinearly coupled with emissivity ϵ_t, surface temperature T_t, and reflected ambient thermal radiation power.



Surface temperature

$$\hat{T}_p\left(\overrightarrow{P_t}; T_t, \epsilon_t\right) = \frac{C_2}{C_3 \ln\left(\frac{C_1}{V_1+V_2}+1\right)} + \frac{C_4}{C_3} \quad (1)$$

Emissivity

Detected Temperature Value

$$V_1 = \frac{GD^2\zeta A_p}{4f^2} \frac{\langle \overrightarrow{P_t}, \vec{x} \rangle^4 e^{-\gamma|\overrightarrow{P_t}|}}{|\overrightarrow{P_t}|^4} \epsilon_t \int_{\lambda_1}^{\lambda_2} M_\lambda^{\mathrm{BB}}(T_t) d\lambda \quad (2)$$

Depth

$$V_2 = \frac{GD^2 A_p \xi_t \zeta}{4f^4} \frac{\langle \overrightarrow{P_t}, \vec{x} \rangle^7 e^{-\gamma|\overrightarrow{P_t}|}}{|\overrightarrow{P_t}|^5} (1-\epsilon_t) \sum_{k=1}^{K} e^{-\gamma|\overrightarrow{P_t}-\overrightarrow{P_i^k}|} S_{\lambda,k} \quad (3)$$

Reflected Thermal Radiation Power

Issue:

Ground truth for emissivity, surface temperature, and reflected radiation is unavailable.

TAP3D Model Design

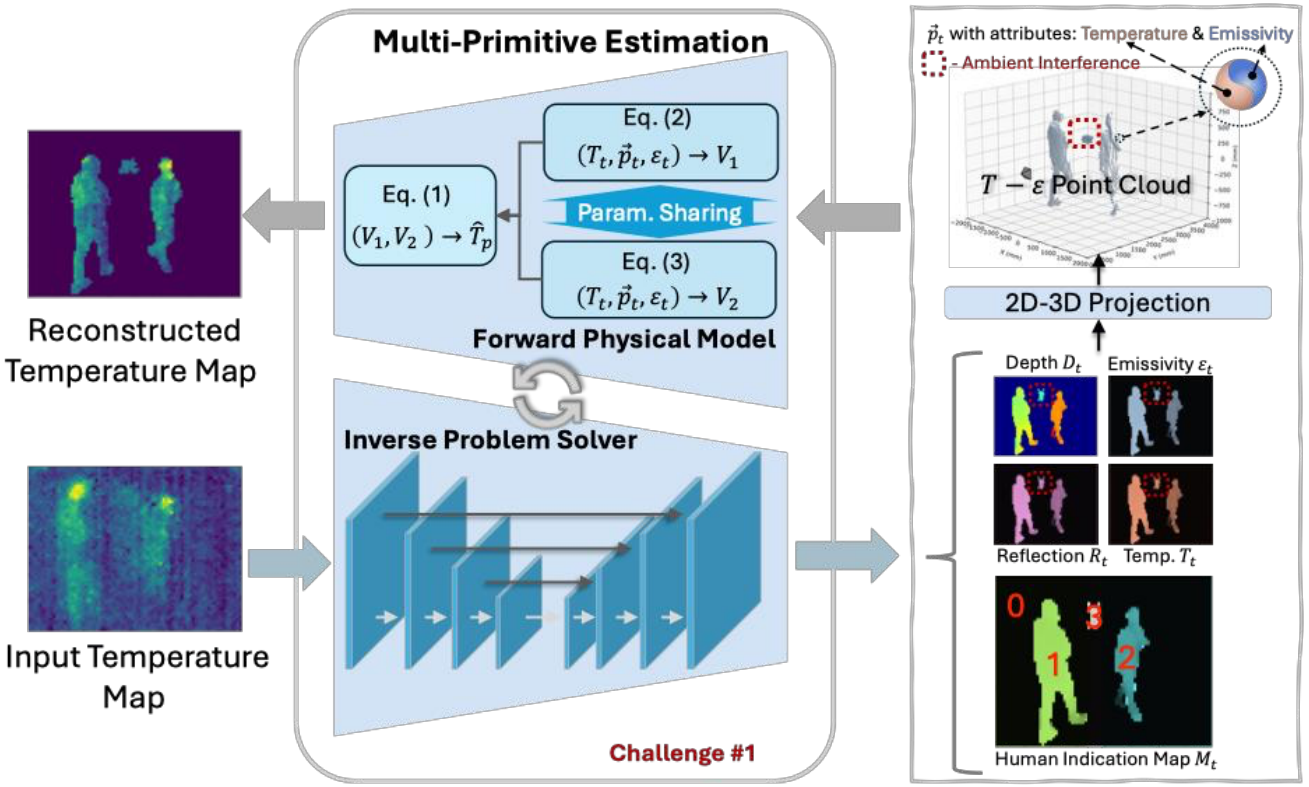
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Thermal interference suppression

Multi-user separation

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Surface temperature

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Emissivity

Detected Temperature Value

$$V_1 = \frac{GD^2\zeta A_p}{4f^2} \frac{\langle \vec{P_t}, \vec{x} \rangle^4 e^{-\gamma|\vec{P_t}|}}{|\vec{P_t}|^4} \epsilon_t \int_{\lambda_1}^{\lambda_2} M_\lambda^{\mathrm{BB}}(T_t) d\lambda \quad (2)$$

Depth

$$V_2 = \frac{GD^2 A_p \xi_t \zeta}{4f^4} \frac{\langle \vec{P_t}, \vec{x} \rangle^7 e^{-\gamma|\vec{P_t}|}}{|\vec{P_t}|^5} (1-\epsilon_t) \sum_{k=1}^{K} e^{-\gamma|\vec{P_t}-\vec{P_i^k}|} S_{\lambda,k} \quad (3)$$

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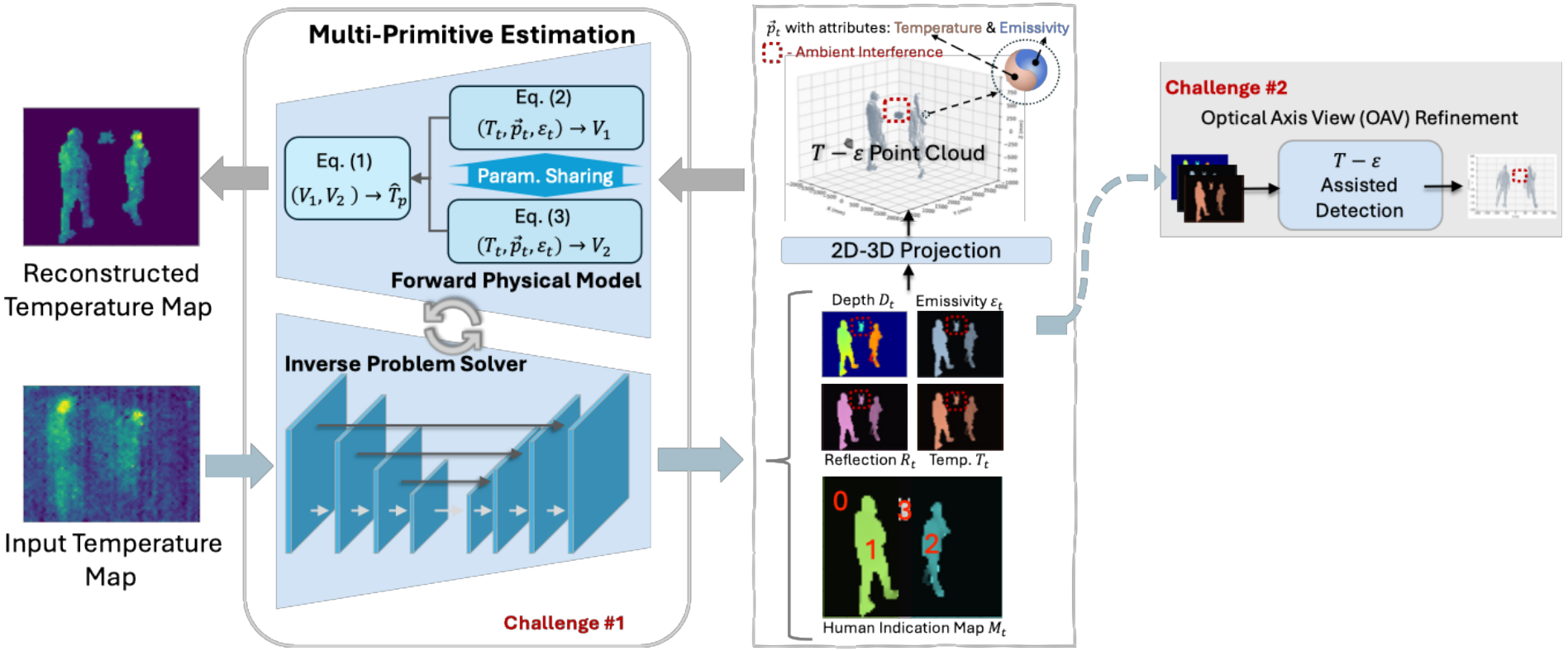
Accurate depth recovery

Thermal interference suppression

Multi-user separation

Key insight:

Surface temperature and emissivity are distinctive features of the human body.



TAP3D Model Design

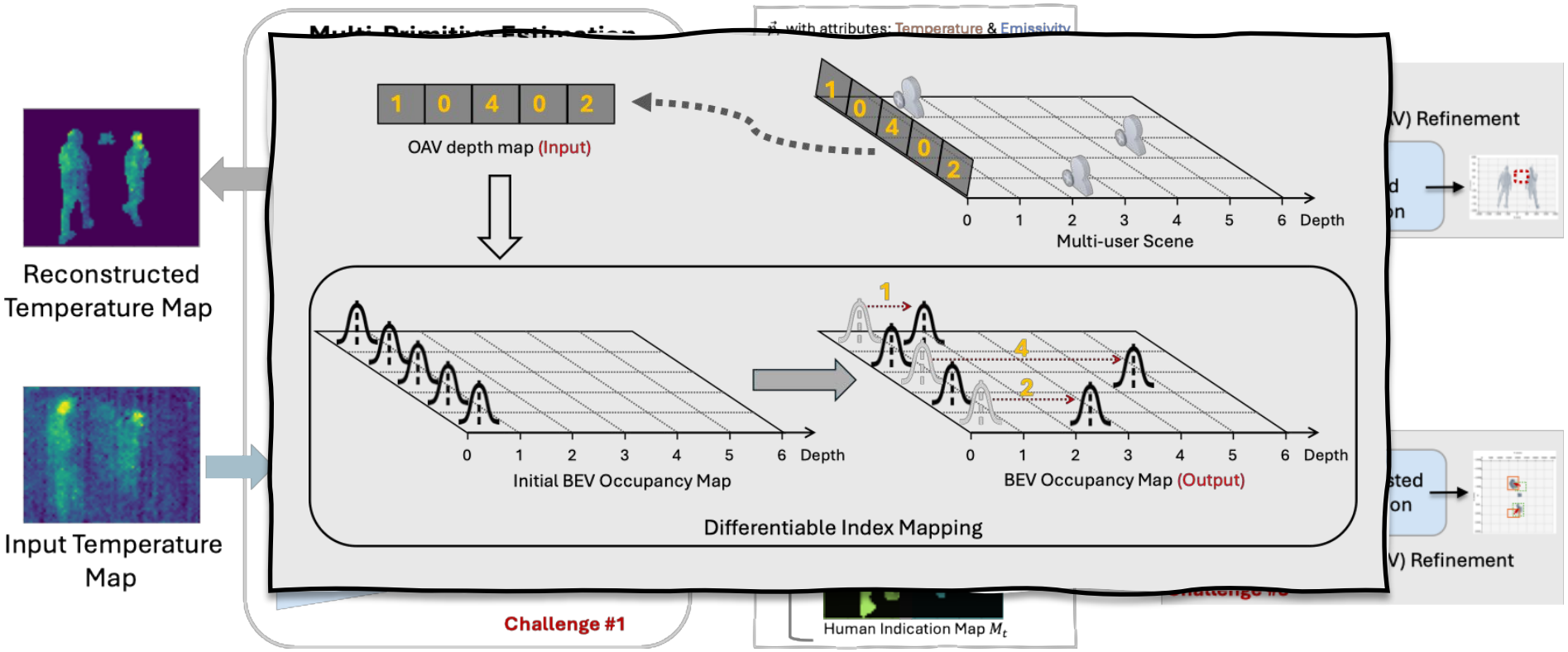
Accurate depth recovery

Thermal interference suppression

Multi-user separation

Key observation:

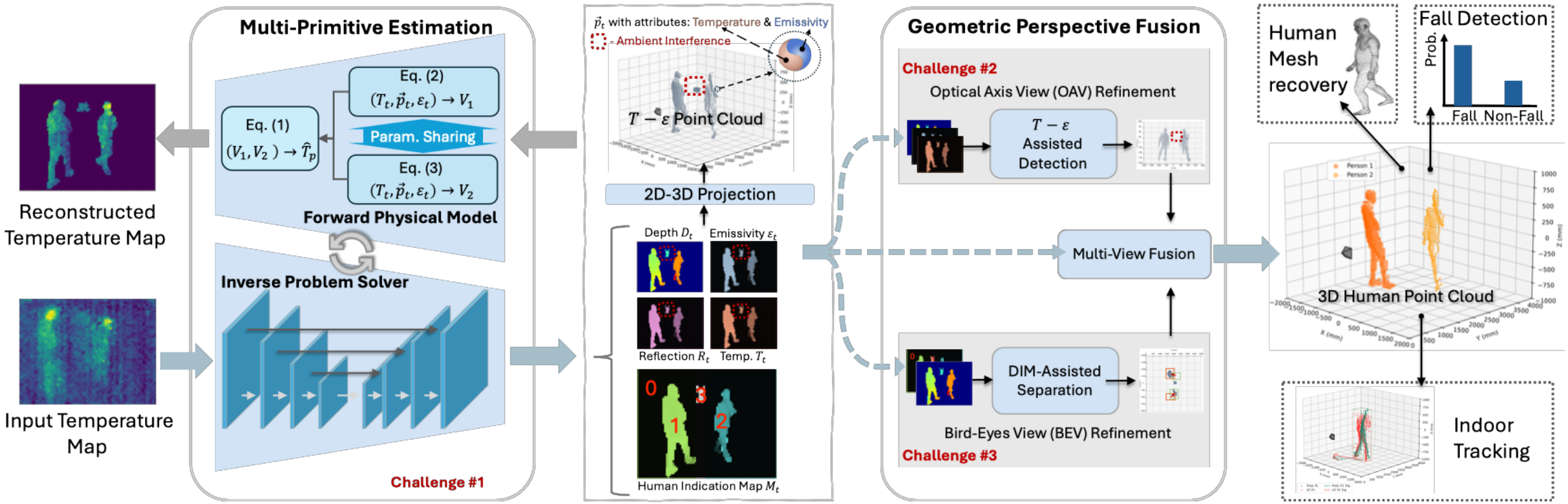
Individuals close or connected in OAV can be more easily distinguished in Bird's Eye View (BEV).



Issue: OAV primitives require a differentiable, invertible transform to BEV for end-to-end training.

Physics-informed TAP3D Model

- Accurate depth recovery ✓
- Thermal interference suppression ✓
- Multi-user separation ✓



TAP3D Evaluation

Dataset:

160K+	8	11	80K
synchronized frames	indoor environments	participants	test samples

Sensor:

$10	80×62	8 Hz	23.3M
thermal array	thermal pixels	capture rate	parameters

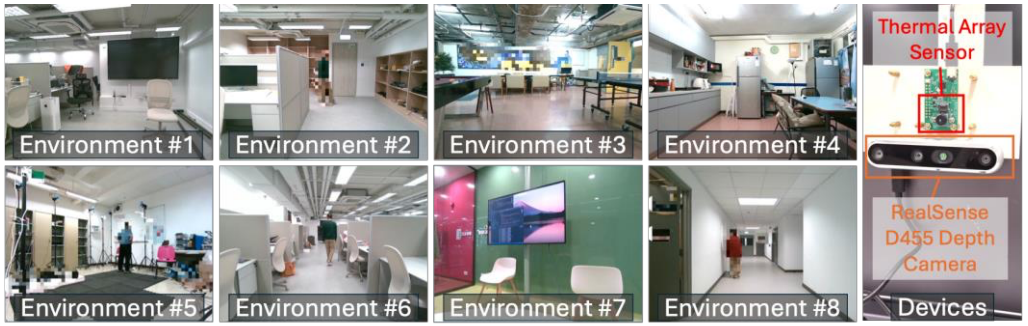


Fig 1: Data collection environments and devices.

Overall Performance:

Tab1: Comparison with SOTA baselines.

Metrics	RGB2Point [22]	NeWCRF [70]	**TAP3D**
DCD (cm)↓	10.15 ± 0.77	19.26 ± 1.50	**4.60 ± 0.25**
F1 Score↑	0.945	0.677	**0.953**
Params (M)↓	176.07	88.46	**23.32**
FLOPs (G)↓	16.96	78.14	**7.41**

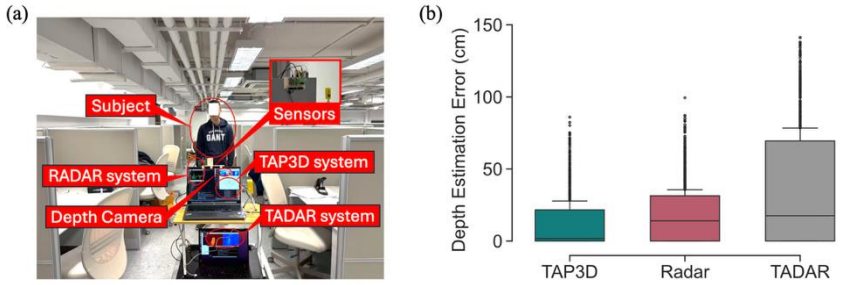
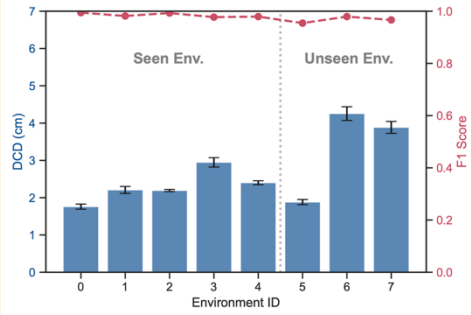


Fig 2: Performance comparison between TAP3D and other low-cost sensing systems.

Detailed robustness gallery

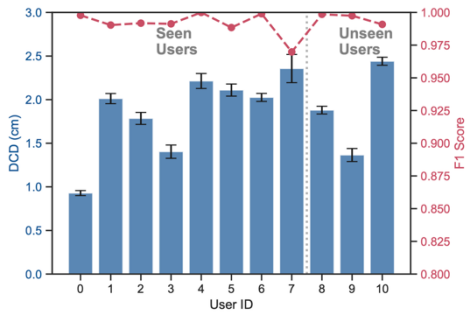
Cross-environments

3.34 cm DCD
0.967 F1



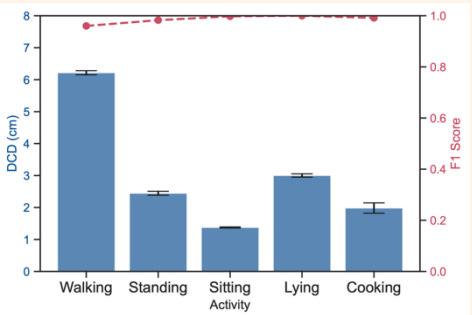
Cross-users

1.90 cm DCD
0.996 F1



Different activities

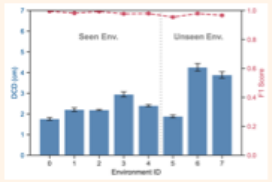
1.4 - 6.4 cm DCD
0.96 -1.00 F1



Detailed robustness gallery

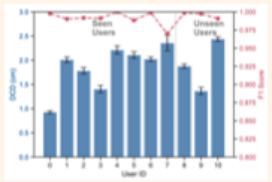
Cross-environments

3.34 cm DCD
0.967 F1



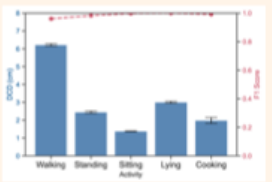
Cross-users

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0.996 F1



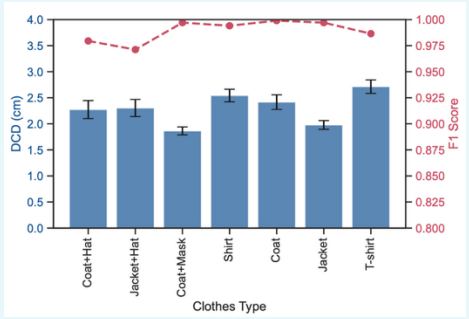
Different activities

1.4 - 6.4 cm DCD
0.96 -1.00 F1



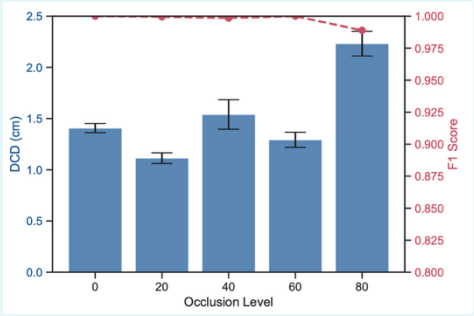
Different Clothing

1.87 – 2.71 cm DCD
> 0.97 F1



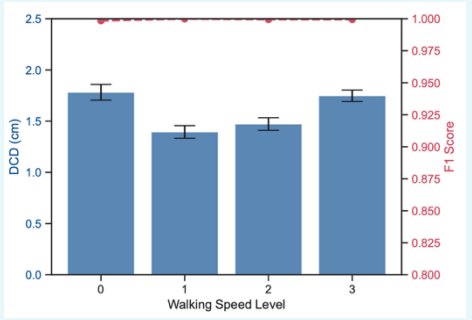
Different occlusion levels

< 2.23 cm DCD
> 0.989 F1



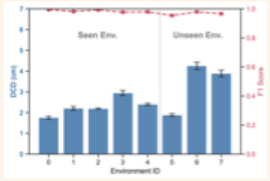
Different walking speed

< 1.8 cm DCD
~ 1.0 F1

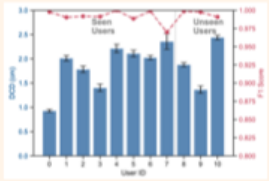


Detailed robustness gallery

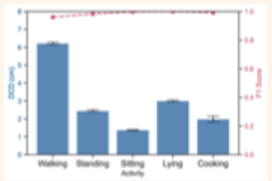
Cross-environments
3.34 cm DCD
0.967 F1



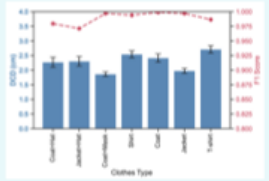
Cross-users
1.90 cm DCD
0.996 F1



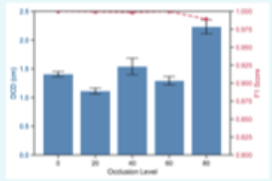
Different activities
1.4 - 6.4 cm DCD
0.96 -1.00 F1



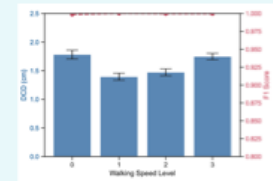
Different Clothing
1.87 – 2.71 cm DCD
> 0.97 F1



Different occlusion levels
< 2.23 cm DCD
> 0.989 F1

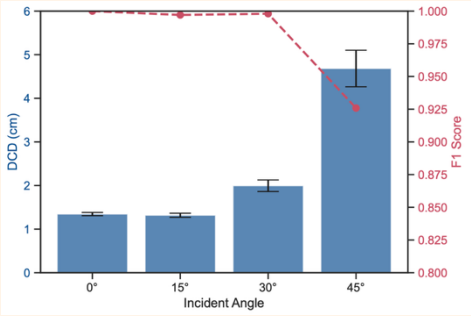


Different walking speed
< 1.8 cm DCD
~ 1.0 F1



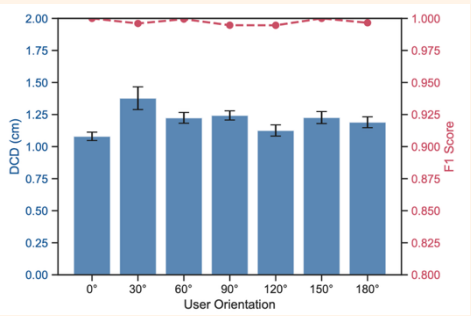
Different Incident angle

< 4.68 cm DCD
> 0.926 F1



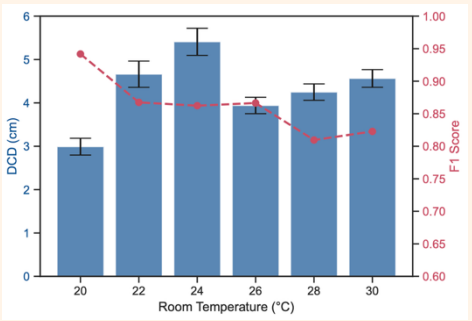
Different user orientations

> 1.38 cm DCD
< 0.995 F1



Room temperatures

< 5.41 cm DCD
> 0.81 F1



TAP3D Evaluation

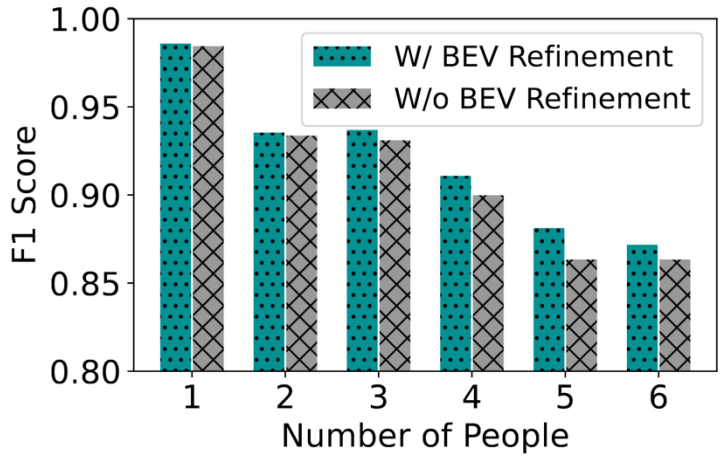


Fig 1: Effect of BEV refinement under varying numbers of users.

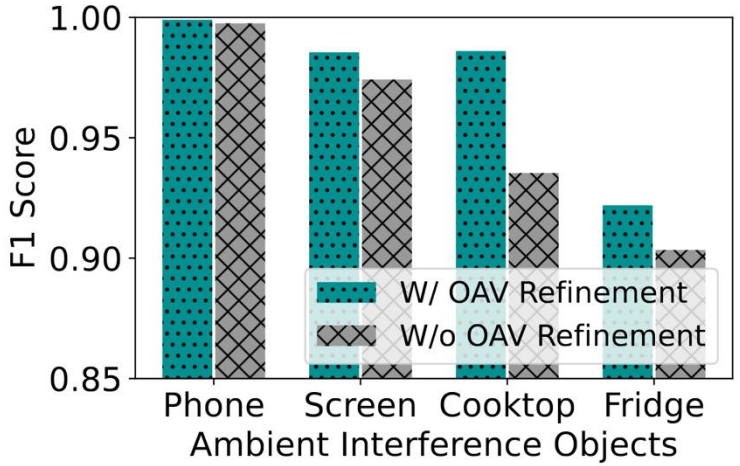


Fig 2: Effect of OAV refinement under different ambient thermal interferences.

BEV refinement is particularly effective in separating multi-user.

Leveraging depth together with emissivity and surface temperature cues effectively reduces false positives caused by non-human heat sources.

TAP3D Evaluation

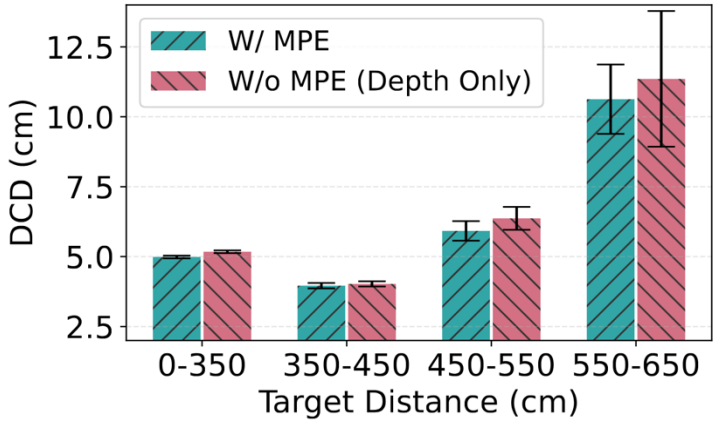


Fig 1: Effect of multi-primitive estimation (MPE) on point cloud reconstruction accuracy.

Multi-primitive estimation reduces DCD by 5.59%, mitigating thermal depth ambiguity.

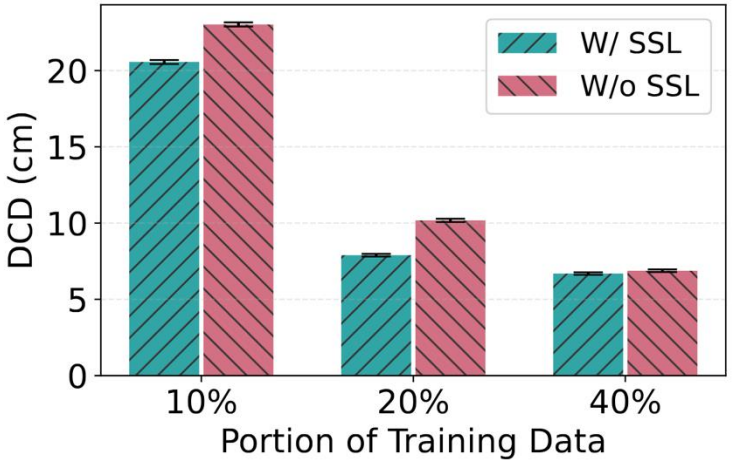


Fig 2: Effect of self- supervised pretraining with different amounts of labeled finetuning data.

Reduces DCD by 11.93%, highlighting the scalability of physics-guided self-supervision for large-scale thermal sensing.

TAP3D Case Study

91.46%

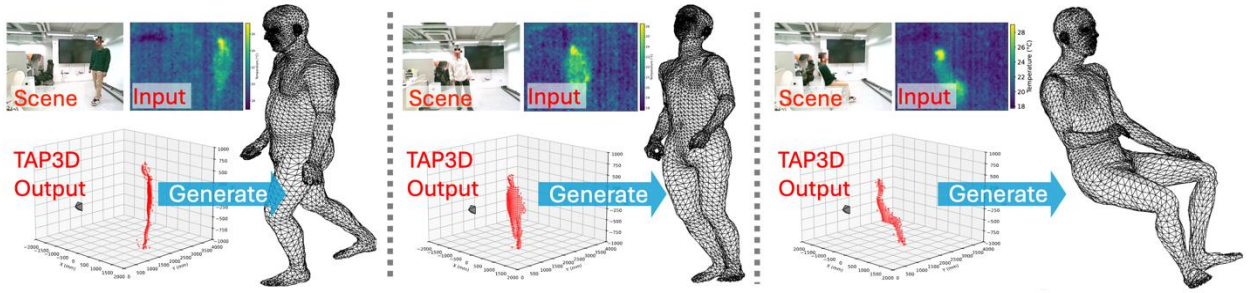
fall detection

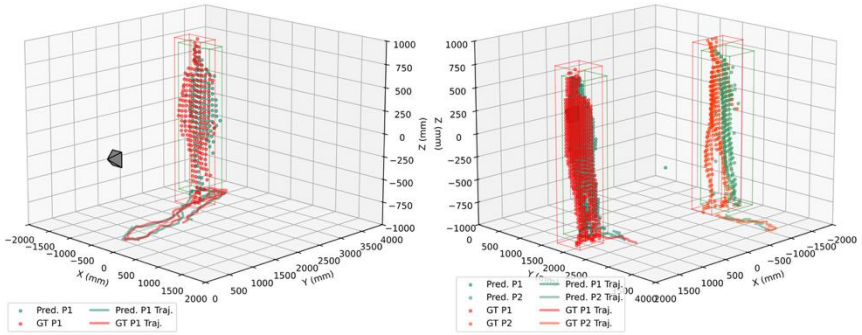
21.86 cm

tracking MAE

4.87 cm

mesh matching

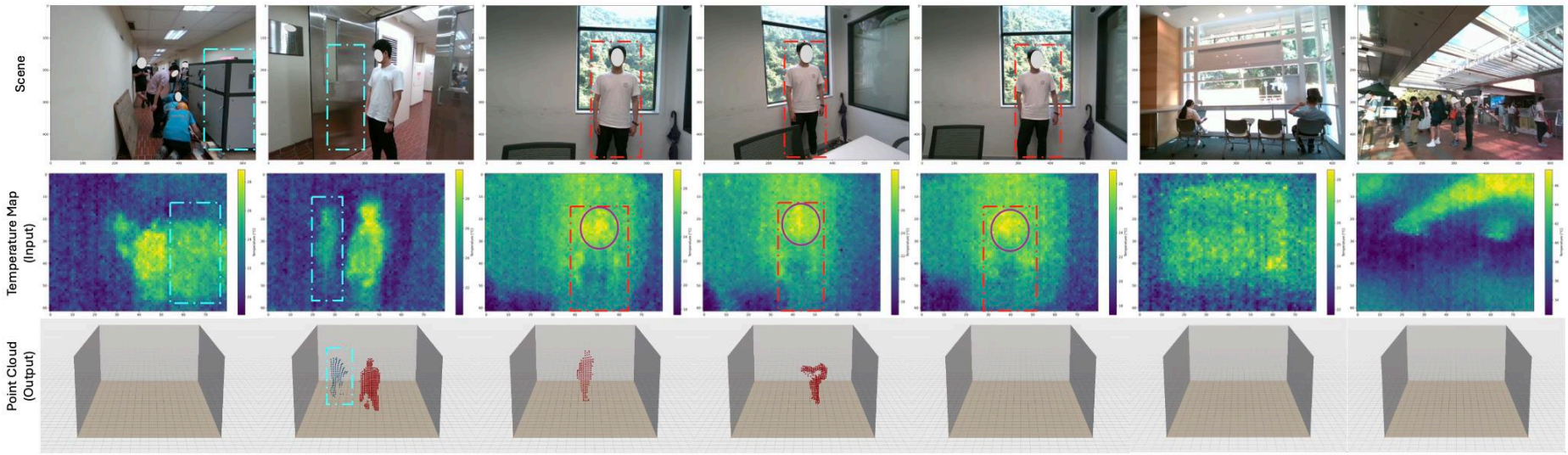




Across 144 one-minute recordings, it achieves a mean center distance error of 21.86 cm

The reconstructed models achieve a minimum matching distance of 4.87 cm.

Limitations and Future Works



Limitations

- **Challenging environments:**
  - Severe multi-user overlap
  - Reflected thermal radiation
  - Sensitive to solar heating
  - Weak human–background contrast
- **Point-cloud density:** Limited by thermal-array resolution
- **Geometric primitives:** Primarily focuses on depth estimation

Future Works

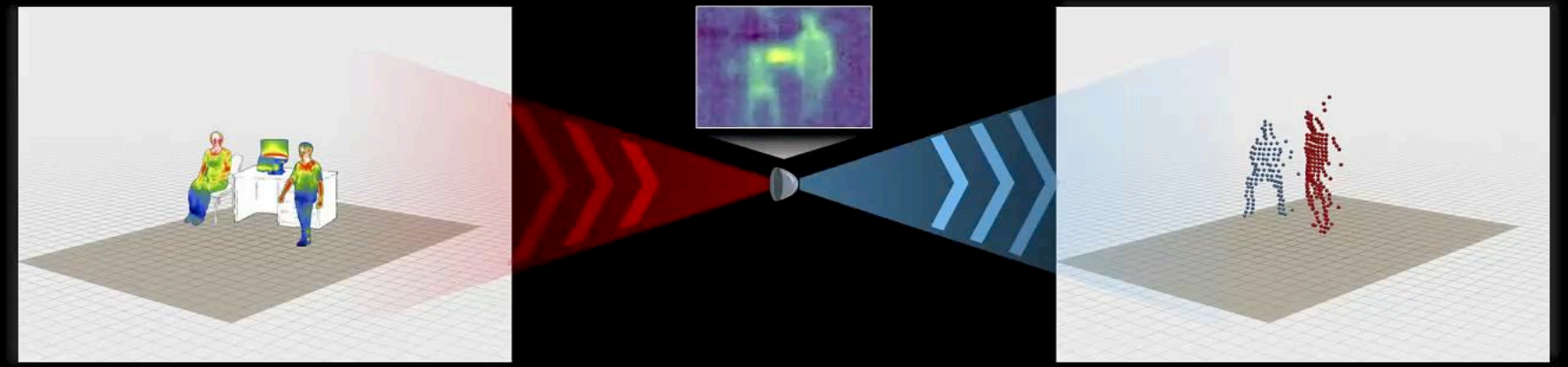
- **Beyond humans:** Extend to non-human targets → Panoptic 3D reconstruction
- **Higher resolution:** Super-resolution / learning-based point-cloud densification
- **Multi-primitive estimation:** Recover richer geometric properties beyond depth
- **Cross-modal fusion:** Align thermal point clouds with LiDAR, radar, and depth cameras

TAP3D

Thermal-Assisted 3D Human Point Clouds

*Xie Zhang, Chengxiao Li, Xuan Liu, Chenshu Wu**

The University of Hong Kong



Scene Under Thermal Vision

Human Point Cloud

HKU AIoT Lab

MobiCom'26

香 港 大 學
THE UNIVERSITY OF HONG KONG

UbiComp ISWC 2026

Tutorial MuMoSe at UbiComp/ISWC, Shanghai, China



RF-LEGO

Modularized Signal Processing–Deep Learning Co-Design for RF Sensing via Deep Unrolling

Luca Jiang-Tao Yu, Chenshu Wu

The University of Hong Kong

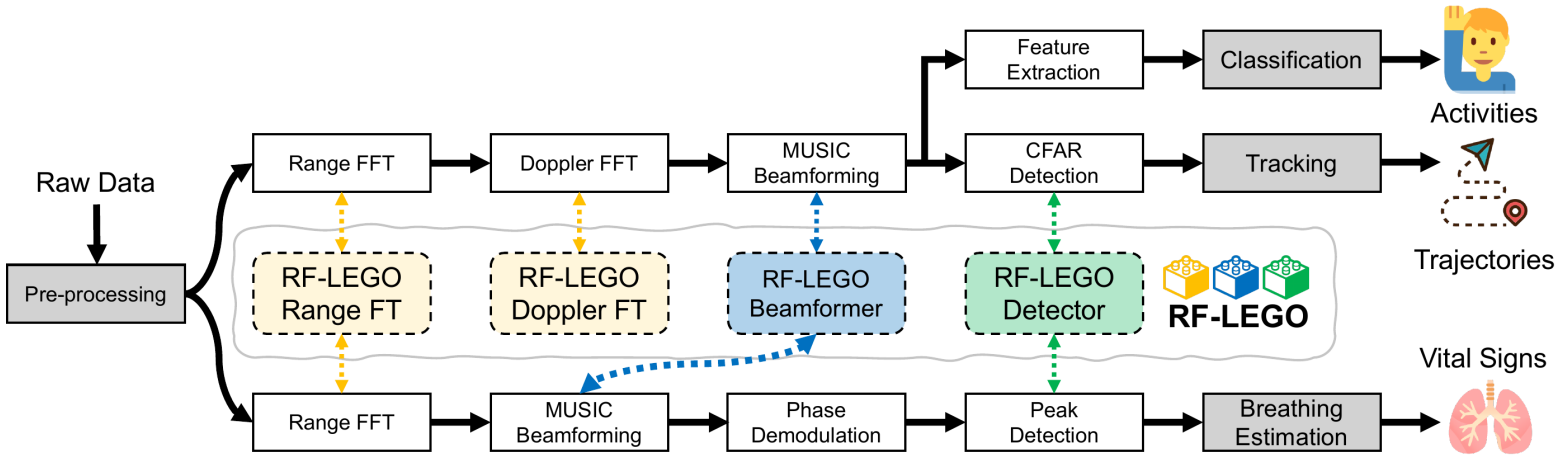


To appear at ACM MobiCom 2026, Austin, TX, USA

Shared signal-processing blocks across various sensing tasks.

Trajectory tracking, vital sign monitoring, and human activity recognition repeatedly ask:

- **At what frequency?**
  Range and motion
- **From which direction?**
  Spatial angle
- **Is a target present?**
  Detection



> **Can we learn these shared building blocks once, then reuse them?**

Luca Yu | lucayu@connect.hku.hk

The gap: reusable physics or adaptable learning?

Classical signal processing

Explicit operators. Meaningful interfaces.

- ✓ Frequency and angle spectra are easy to inspect.
- ✓ Blocks cascade into many sensing pipelines.
- ✗ Fixed coefficients need manual tuning.

Task-specific deep learning

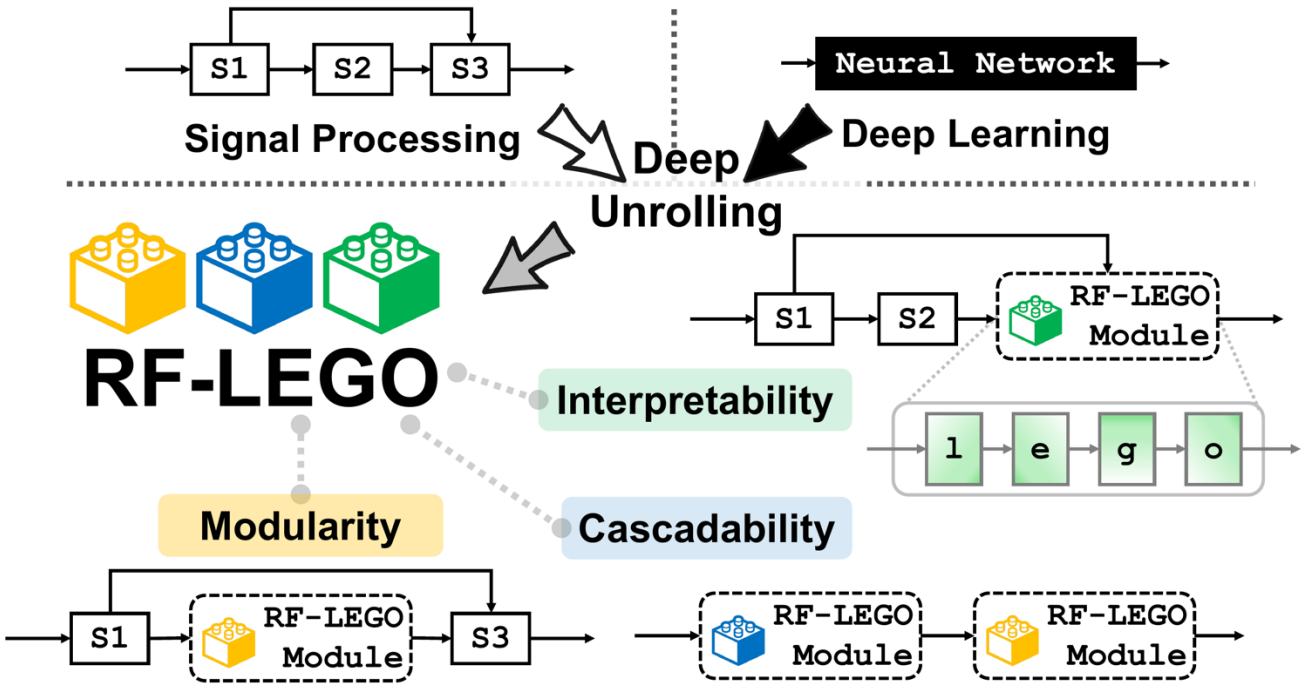
Learned adaptation. Entangled representations.

- ✓ Adapts to complex measurements.
- ✗ Often optimized end to end for one task.
- ✗ Internal features can be hard to reuse.

> **We want adaptation inside a reusable signal-processing interface.**

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Keep the processing structure. Learn the rigid parts.



Modularity

Replace one classical operator.

Cascadability

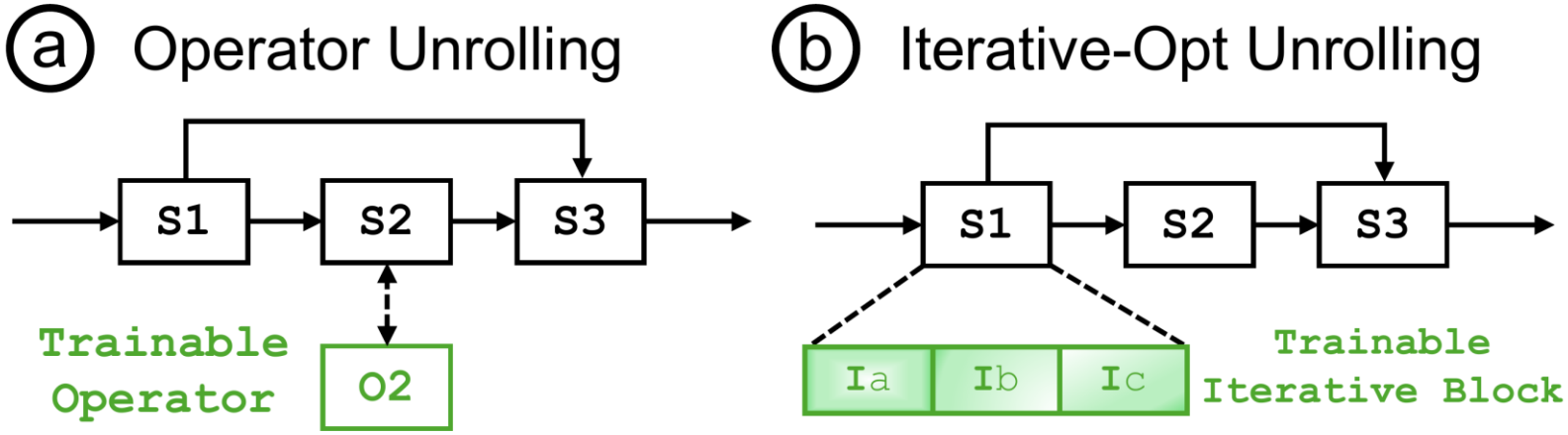
Connect learned and classical blocks.

Interpretability

Retain meaningful operators and outputs.

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Primer on Deep Unrolling



Unroll an operator

Expose a trainable component inside its mathematical structure.

Frequency transform | Detector

Unroll an iterative solver

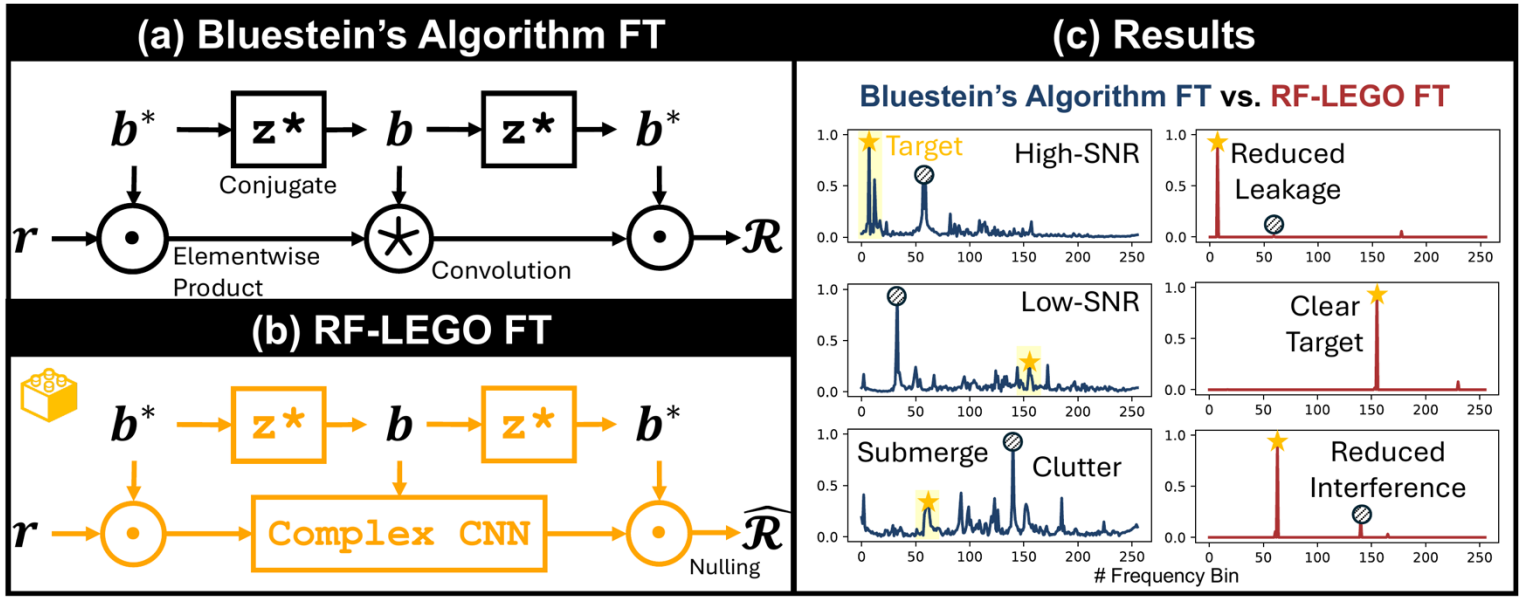
Turn solver iterations into layers with learned updates.

Beamformer

Luca Yu | lucayu@connect.hku.hk

FT: learn the convolution inside a Fourier transform.

Bluestein's algorithm exposes a convolution that can adapt to signal artifacts.



Preserve

Outer chirp multiplications and complex-valued signals.

Learn

A shallow complex convolution, initialized from the chirp.

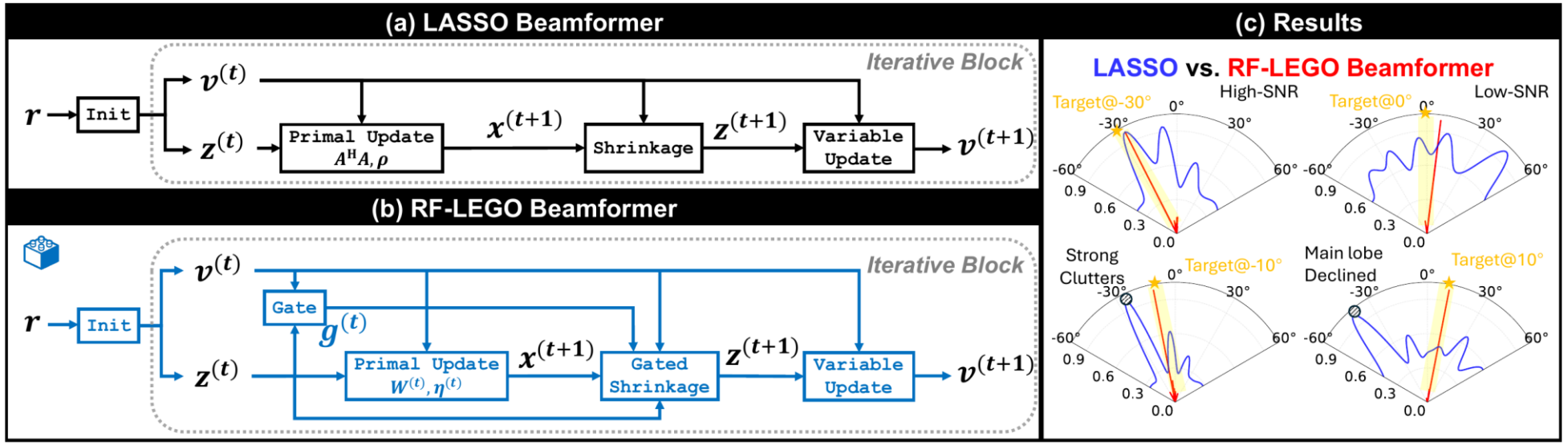
Improve

Suppress spectral leakage and reveal weak targets.

Luca Yu | lucayu@connect.hku.hk

Beamformer: learn how an angular spectrum is refined.

Unroll a sparse LASSO estimator into ten trainable ADMM stages.



Preserve

Steering model and explicit angle spectrum.

Learn

Update coefficients and gates between iterations.

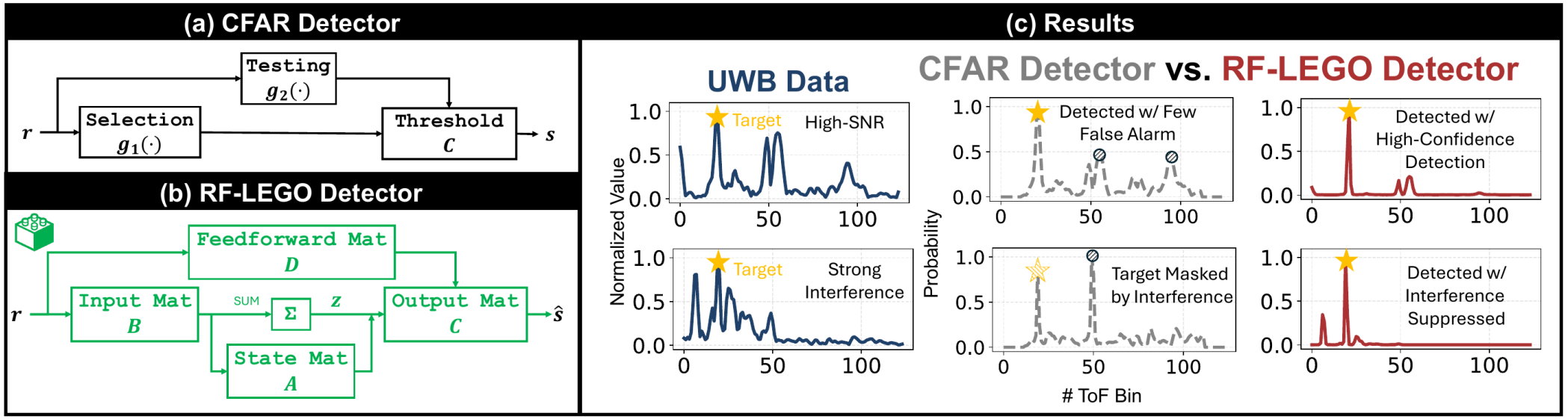
Avoid

Dependence on a subspace eigendecomposition.

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Detector: retain selection and testing, then add memory.

Replace rigid local thresholding with a trainable state-space operator.



Preserve

The selection-and-testing structure.

Learn

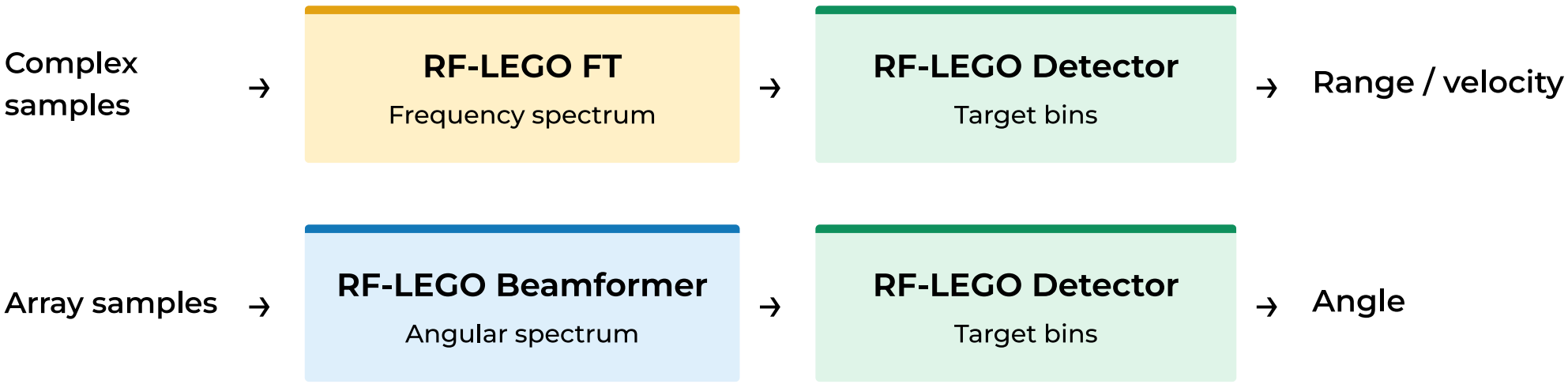
State evolution and input/output mappings.

Control

A detection operating point at a matched false-alarm rate.

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Cascadability



Swap one block

Isolate its contribution in the same pipeline.

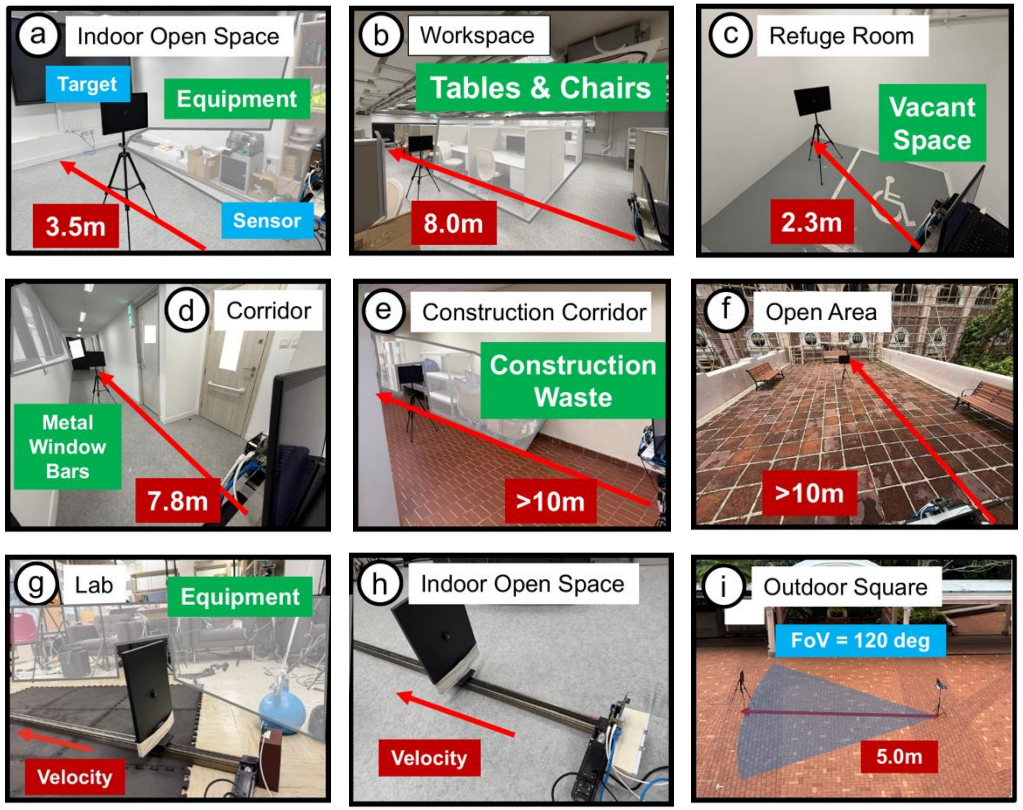
Cascade several blocks

Feed a classical estimator or a neural back end.

> **Cascadability comes from the signal representation between blocks.**

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Train on synthetic signals. Evaluate on real measurements.



Devices & datasets

mmWave radar · UWB radar · Wi-Fi

UWCR — mmWave radar
OPERAnet — Wi-Fi CSI
UWB-Context — UWB CIR / ToF
DeepSense 6G — 60 GHz mmWave, beam selection

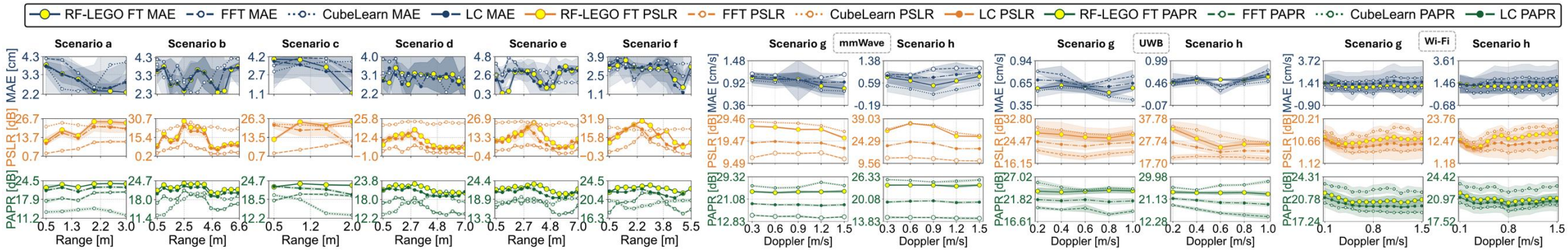
Baselines

- Signal Processing Baselines
- Deep Learning Baselines
- Loose Coupling Baselines

Data **30k** synthetic frames per module **≈3M** real-world frames

Luca Yu | lucayu@connect.hku.hk

FT reduces estimation error across sensing modalities.



Signal Processing Baselines

≈10% lower range MAE
≈20% lower Doppler MAE
than FFT.

Deep Learning Baselines

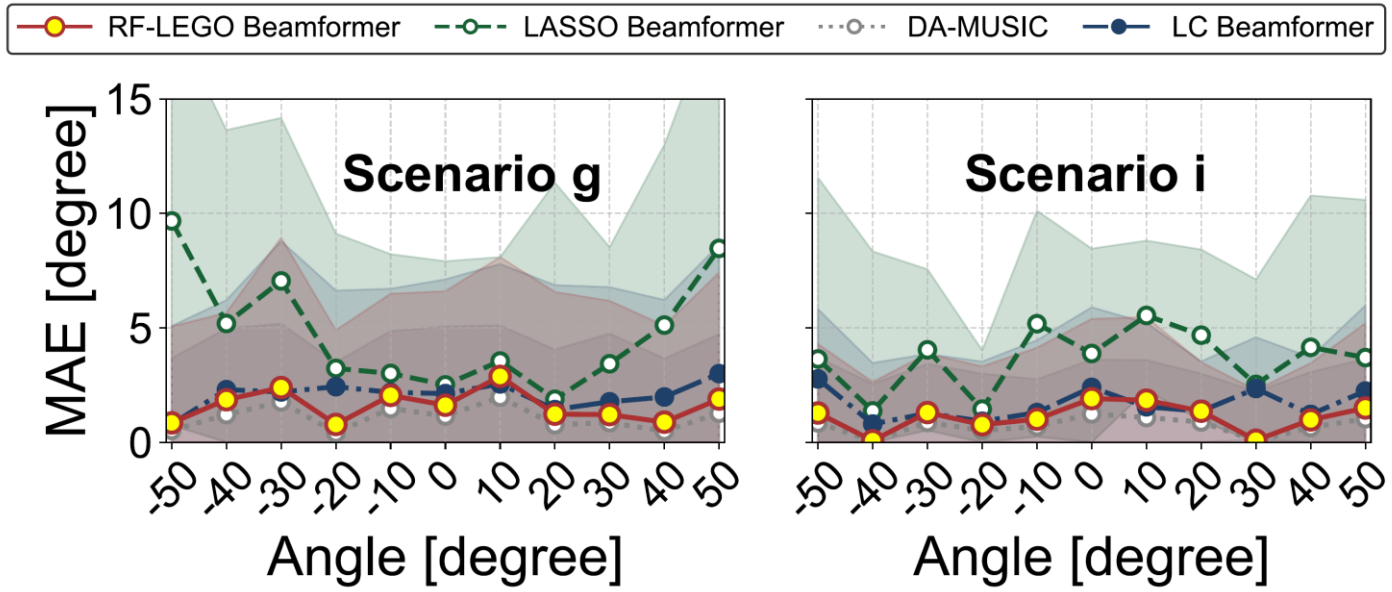
Comparable accuracy to
the adapted **CubeLearn** front end.

Loose Coupling Baselines

Outperforms the parameter-matched **LC FT**.

Luca Yu | lucayu@connect.hku.hk

Beamformer cuts angular error by 68%.



Signal Processing Baselines

4.23° → 1.35° MAE
68% lower than the LASSO beamformer.

Deep Learning Baselines

Competitive with **DA-MUSIC**, without a subspace eigendecomposition.

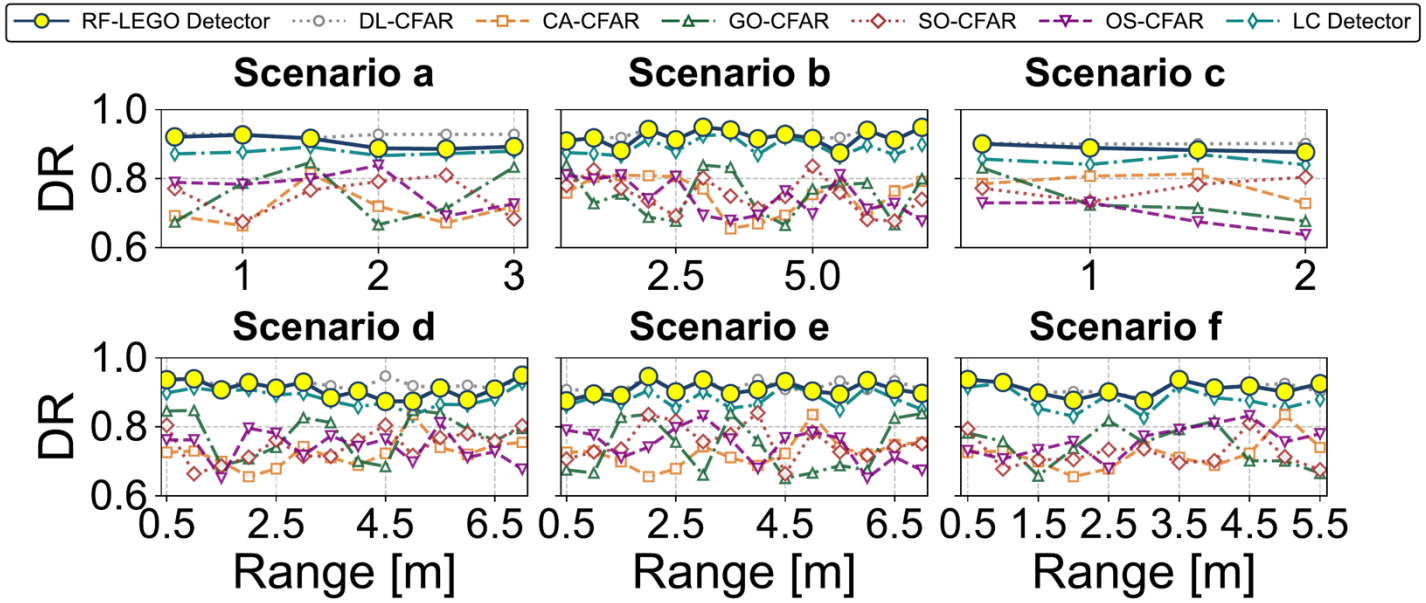
Loose Coupling Baselines

Lower angular MAE than the **LC Beamformer**.

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Detector improves detection at a matched false-alarm rate.

False-alarm rate **at 10^{-3} order of magnitude**



Signal Processing Baselines

Higher detection rate than **CA-CFAR, GO-CFAR, SO-CFAR, and OS-CFAR**.

Deep Learning Baselines

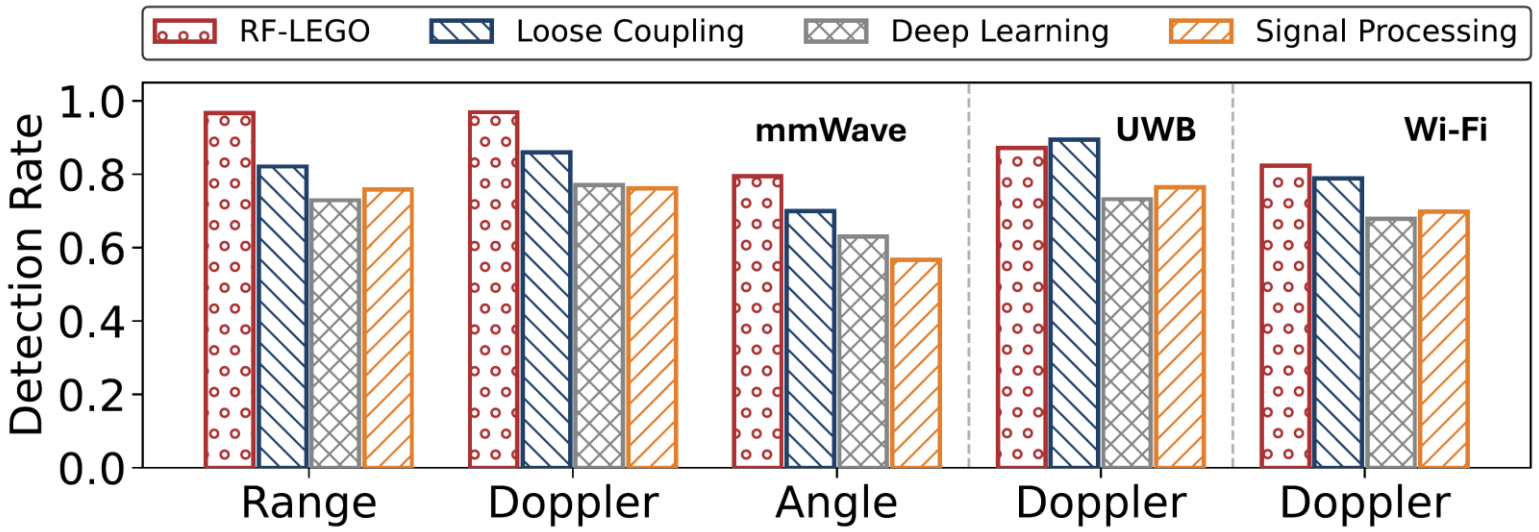
Competitive detection rate with **DL-CFAR**.

Loose Coupling Baselines

On par with the **LC Detector** across the UWB scenarios.

Luca Yu | lucayu@connect.hku.hk

Cascadability across sensing modalities



Relative DR gains over Signal Processing Baselines on mmWave at FAR = 10^{-3}

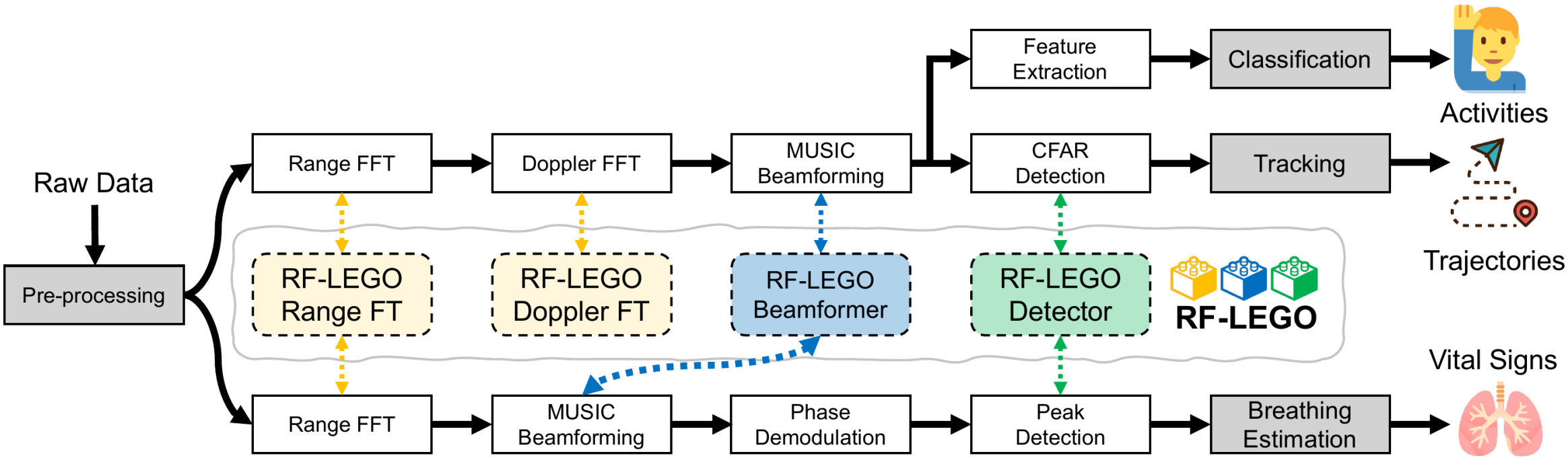
+27.6% Range

+27.3% Doppler

+40.2% Angle

Luca Yu | lucayu@connect.hku.hk

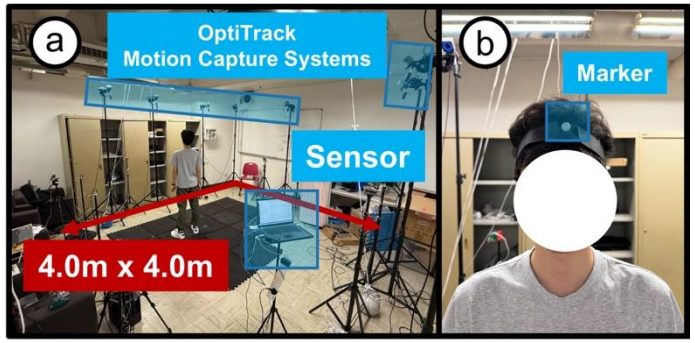
Case Study Pipelines



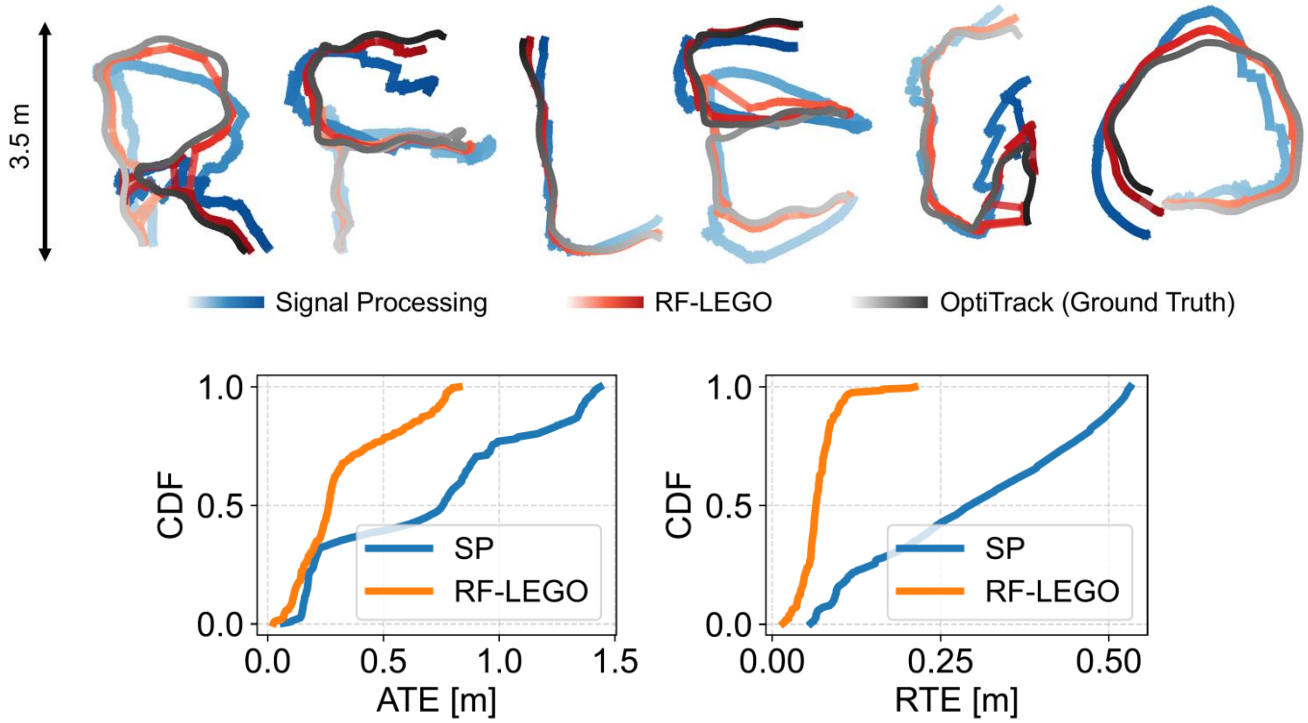
Trajectory Tracking **Vital Sign Monitoring** **Human Activity Recognition**

Luca Yu | lucayu@connect.hku.hk

Trajectory Tracking



4 m × 4 m room · OptiTrack
65 trajectories across 13 categories



Median trajectory error

≈0.3 m

Absolute trajectory error (ATE)

Improvement over SP

≈40% lower

Median absolute trajectory error

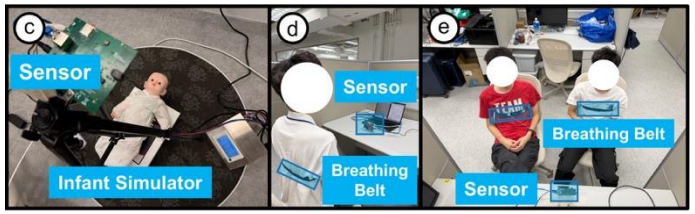
Downstream tracker

Same EKF

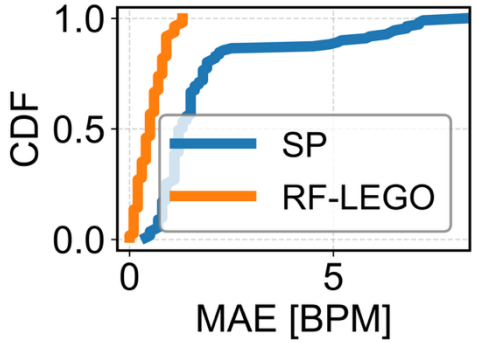
Replace FFT / MUSIC / CFAR

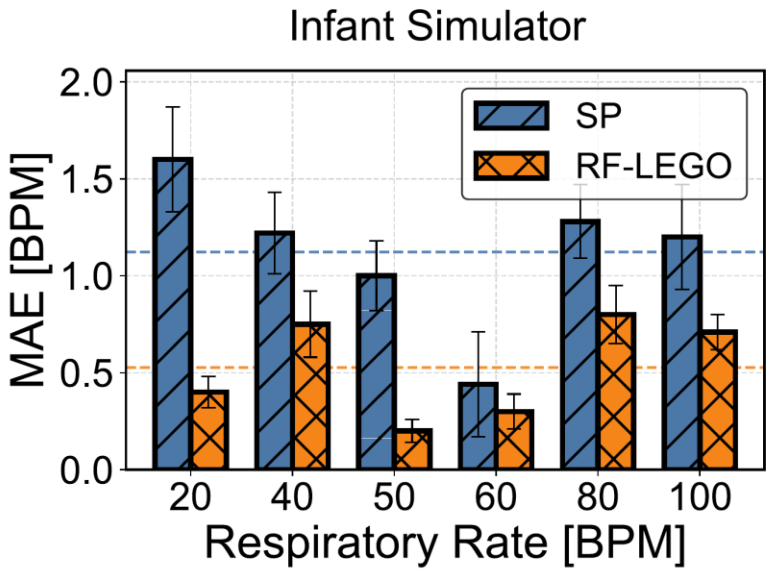
Luca Yu | lucayu@connect.hku.hk

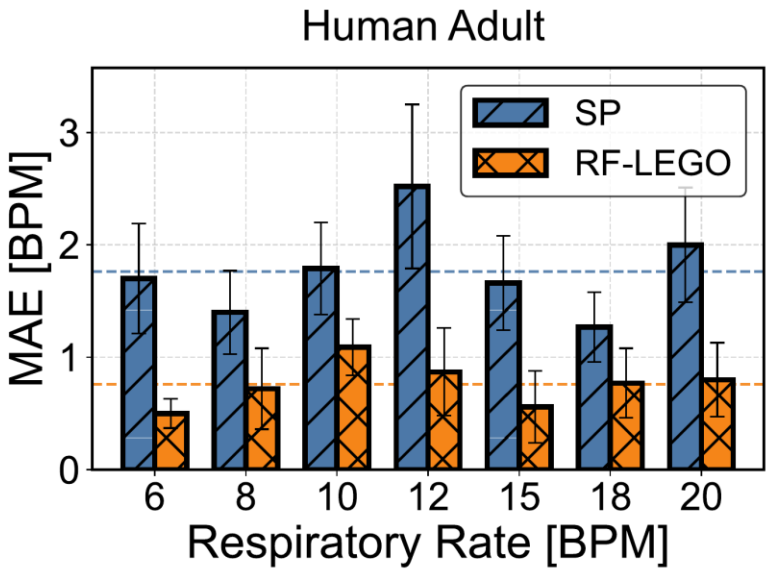
Vital Sign Monitoring



Infant simulator · Human adults
Single-person and multi-person settings







Infant simulator

52.7% lower

Breathing-rate MAE vs. SP

Human adults

56.8% lower

Breathing-rate MAE vs. SP

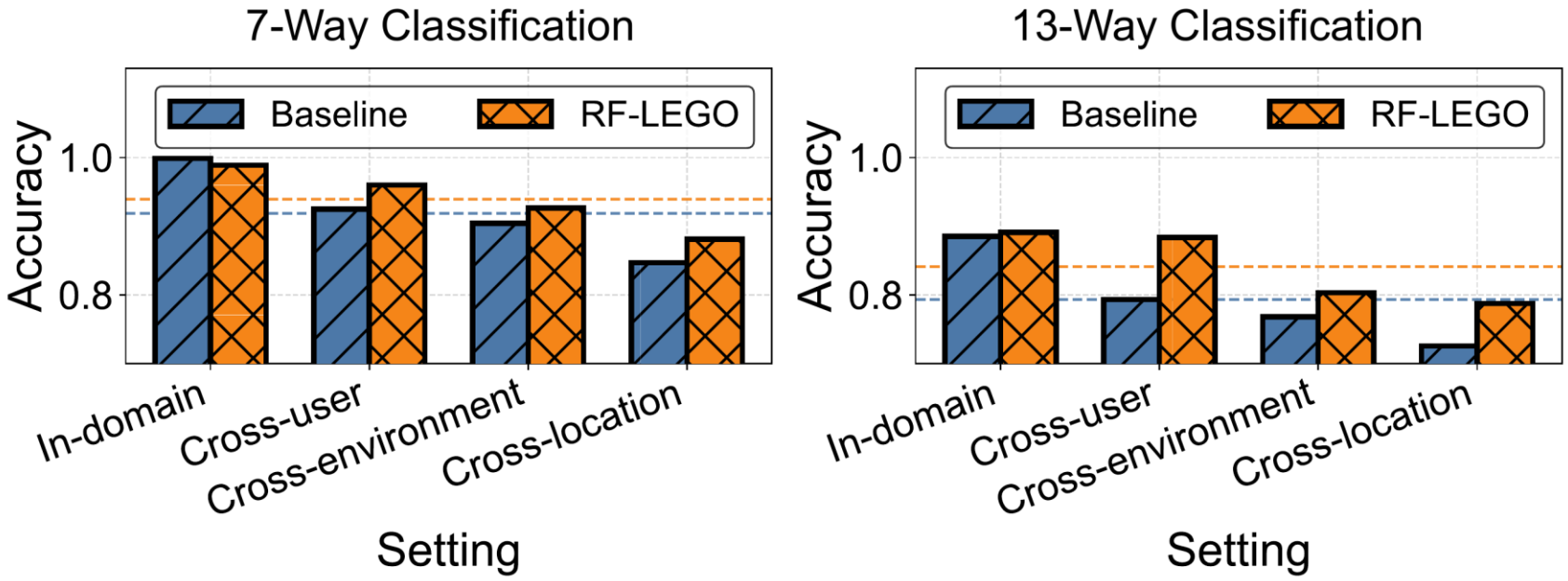
Downstream processing

Same estimator

Retain phase analysis and rate estimation

Luca Yu | lucayu@connect.hku.hk

Human Activity Recognition



MCD-Gesture · mmWave radar · 6 environments · 25 users · 5 locations

7-way recognition

91.88% → 93.92%

Mean accuracy across four splits

13-way recognition

79.31% → 84.15%

Mean accuracy across four splits

Neural back end

Same backbone

Only the RF front end changes

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Accuracy comes with measurable computation overhead.

Device	RF-LEGO FT	RF-LEGO Beamformer	RF-LEGO Detector
Jetson Orin Nano	**0.76** / 0.03	**2.60** / 0.73	**6.20** / 1.74
Raspberry Pi 4	**3.08** / 0.17	**8.33** / 1.93	**30.86** / 4.59
ESP32-P4	**31.07** / 0.36	**81.68** / 3.80	**122.46** / 7.76

RF-LEGO / Signal Processing Baselines, in milliseconds. Individual operators.

Small edge computer

Millisecond-scale modules on Orin Nano.

Microcontroller

Latency remains tolerable for sensing tasks with relaxed update-rate requirements.

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Takeaways of RF-LEGO

Learn the operator.

Keep the interface.

Cascade the modules.

Three reusable RF blocks

Frequency transform · Beamformer · Detector

Verified in complete sensing pipelines

Trajectory Tracking
Vital Sign Monitoring
Human Activity Recognition

MobiCom 2026 Artifact Evaluation



Artifacts Available



Artifacts Evaluated
Functional



Artifacts Evaluated
Reusable



Results Reproduced

Luca Yu | lucayu@connect.hku.hk

Thank you!

Project

aiot-lab.github.io/RF-LEGO

Code

github.com/aiot-lab/RF-LEGO

Homepage

1ucayu.github.io

Slides

Scan to open these slides

I am actively looking for internship opportunities.

Luca Yu | lucayu@connect.hku.hk